

Local-Unitary Rigidity and Quantitative Rounding for Stabilizer AME States

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Abstract

Let $q = p^e$ and $m \geq 2$. Every product-unitary intertwiner between two stabilizer $\text{AME}(2m, q)$ states is Clifford on each party. The proof recovers the complete local Weyl frame from any $(m + 1)$ -party marginal: the supported stabilizer labels form a q^2 -element group and project bijectively onto the local Weyl labels. These support bijections also form a minimum-support atlas which classifies local-unitary equivalence up to local symplectic changes of frame. As an encoder consequence, every transversal conversion between the associated $[[2m - 1, 1, m]]_q$ stabilizer codes is Clifford on every physical and logical factor.

We prove a quantitative symmetry counterpart for one fixed state without reading the exponentially small Weyl signal of one marginal. Viewing each party as the logical leg of the associated code, a leakage-aware three-region cleaning argument and Weyl–Fourier concentration round every local factor within 8ε of a Clifford. The uniform stabilizer overlap gap then selects an exact symmetry and gives a defect-only decomposition radius

$$\Theta\left(\min\{p^{-1}, q^{-1/2}, n^{-1/2}\}\right), \quad n = 2m,$$

with residual generator norm at most $\pi\sqrt{q}\varepsilon$. At a dimension-only radius the rounded symplectic maps already satisfy the exact minimum-support atlas. The remaining obstruction is affine: localized commutators cancel the stabilizer-character phases and therefore cannot control the product-Pauli correction. On generalized and extended Reed–Solomon AME families the certified scale is $\Theta(q^{-1})$ over prime fields and $\Theta(q^{-1/2})$ at extension degree at least two.

Index Terms—Absolutely maximally entangled states, local Clifford group, local-unitary equivalence, quantitative rigidity, quantum MDS codes, stabilizer cleaning, transversal gates, Weyl operators.

1 Introduction

An absolutely maximally entangled state has maximally mixed reductions on every subsystem no larger than half the parties. Fix a prime power $q = p^e$ and the standard additive Weyl system on $\mathbb{C}^q$. We ask how much of those local Weyl coordinates survives an arbitrary single-party change of basis.

Local-unitary (LU) equivalence does not imply local-Clifford (LC) equivalence for general stabilizer states [JCWY10]. Maximal entanglement changes the answer completely.

Theorem 1.1 (Local-unitary rigidity). Let $q = p^e$, let $m \geq 2$, and let $|\psi\rangle, |\phi\rangle$ be stabilizer $\text{AME}(2m, q)$ states for the standard additive Weyl system. Suppose that, for a party permutation π , single-qudit unitaries $U_1, \dots, U_{2m}$, and a phase θ ,

$$(U_1 \otimes \dots \otimes U_{2m})\pi|\psi\rangle = e^{i\theta}|\phi\rangle.$$

Then every U_i normalizes the Weyl system. Thus the displayed equivalence is local Clifford.

The proof reads one $(m+1)$ -party reduced operator. Stabilizer labels supported there form a group of order q^2 , and projection to the label space at any retained party is bijective. The nonidentity part of the marginal is therefore diagonal on the complete local Weyl basis, with no missing axis. Rank-one contractions recover those axes intrinsically, so product conjugation must permute them. The argument is independent of CSS form, equal phases, classical linearity, and $\mathbb{F}_q$ -linearity of the stabilizer labels:

$$\text{AME support squeeze} \implies \text{full-Weyl marginal} \implies \text{intrinsic axes} \implies \text{Clifford factors.} \quad (1.1)$$

The approximate fixed-state symmetry problem cannot use this proof naively. The non-identity part of one minimum-support marginal has exponentially small Hilbert–Schmidt norm. Instead, choose one tensor leg as the logical input of the associated $[[2m-1, 1, m]]_q$ code. Its physical parties split into three correctable regions. A nested commutator localizes on one region, scalar compression controls leakage, and Weyl–Fourier concentration recovers a Clifford on the logical leg. Repeating this with each party as input gives the following global statement.

Theorem 1.2 (Quantitative rounding). Let $|\psi\rangle$ be a stabilizer $\text{AME}(n, q)$ state with $n = 2m$, $m \geq 2$, and let $U = \bigotimes_i U_i$. Write

$$\varepsilon(U) = \min_{|z|=1} \|U|\psi\rangle - z|\psi\rangle\|$$

and put

$$\begin{aligned} \tau_p &= \min \left\{ \frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}} \right\}, & d_p &= (2 - 2p^{-1/2})^{1/2}, \\ R_{n,q}^{\text{clean}} &= \min \left\{ \frac{\tau_p}{8}, \frac{1}{4\sqrt{2}q}, \frac{1}{8\pi\sqrt{n}}, \frac{d_p}{1 + 4\pi\sqrt{n}} \right\}. \end{aligned} \quad (1.2)$$

If $\varepsilon(U) < R_{n,q}^{\text{clean}}$, then, up to phase,

$$U = g \bigotimes_{i=1}^n e^{ih_i}, \quad g \in G(\psi), \quad \left(\sum_i \|h_i\|_F^2 \right)^{1/2} \leq \pi\sqrt{q}\varepsilon(U), \quad (1.3)$$

where the h_i are traceless Hermitian with spectral spread at most π . Before the exact branch is selected, each U_i is within normalized Hilbert–Schmidt distance $8\varepsilon(U)$ of a one-qudit Clifford. Uniformly over prime powers and existing AME states,

$$R_{n,q}^{\text{clean}} = \Theta\left(\min\{p^{-1}, q^{-1/2}, n^{-1/2}\}\right).$$

The exact three-cleanable-region Clifford obstruction is due to Pastawski and Yoshida [PY15, Lemma 5], building on stabilizer cleaning [BT09, Lemma 1] and the localization method behind transversal-gate hierarchy bounds [BK13, JOKY18]. The paper makes this mechanism quantitative: an implementation with leakage ε puts every nested logical Weyl commutator within 8ε of a scalar; a finite Fourier argument then rounds it to a Clifford. Quantized overlaps of stabilizer states separate distinct product-Clifford branches by d_p , after which the AME second moment gives the collective residual bound in (1.3). The theorem is a certified lower bound on the rounding radius, not an optimality claim. It concerns approximate symmetries of one fixed state, not arbitrary approximate intertwiners between two states. The latter are treated in Proposition B.21 at the weaker single-marginal scale $q^{-(m+1)/2}$.

The support bijections retain more than the individual Weyl axes. For an $(m+1)$ -party support A , projection at party i gives a bijection $p_i : L_\psi(A) \rightarrow \mathbb{F}_q^2$. The transition maps $M_A(i, j) = p_j p_i^{-1}$ form the *minimum-support atlas*. Their holonomies record how the recovered frames fit together.

Theorem 1.3 (Minimum-support atlas classification). After a fixed party relabelling, two stabilizer $\text{AME}(2m, q)$ states are LU equivalent exactly when their minimum-support atlases are equivalent by local trace-symplectic changes of Weyl frame. For one state, if $\mathcal{G}_\psi$ is the group of compatible local symplectic gauges and Γ_ψ its fixed-party projective product-unitary symmetry group, then

$$1 \longrightarrow L_\psi \longrightarrow \Gamma_\psi \longrightarrow \mathcal{G}_\psi \longrightarrow 1$$

is exact. When q is prime, $\mathcal{G}_\psi$ is the centralizer in $\text{SL}_2(q)$ of the atlas holonomies after base transports are chosen.

The stabilizer hypothesis is essential to this discrete atlas. General AME families can contain infinitely many LU classes; Ramadas and Lakshminarayan give an orthogonal-array criterion producing such families [RL25, Theorem 1 and Corollary 1].

At the dimension-only radius

$$\min \left\{ \frac{\tau_p}{8}, \frac{2 \sin(\pi/p)}{4 + 64\sqrt{q}} \right\}, \quad (1.4)$$

the symplectic maps recovered by cleaning already satisfy every exact atlas transition. This still does not select the exact affine branch. Localized commutators cancel Weyl-translation phases, so they see the stabilizer label space but not its character. The resulting exact product-Pauli correction need not be locally close to the rounded product Clifford. This affine-character obstruction is the precise missing input between robust linear compatibility and a party-count-independent global entry radius.

The encoder consequence of exact rigidity is immediate and operational. Throughout, *transversal* means a tensor product of one-site physical operators with party labels fixed.

Corollary 1.4 (Factorwise transversal rigidity). Let $V_\psi, V_\phi : \mathbb{C}^q \rightarrow (\mathbb{C}^q)^{\otimes(2m-1)}$ be encoders whose normalized Choi states are stabilizer $\text{AME}(2m, q)$ states. If

$$(U_1 \otimes \cdots \otimes U_{2m-1})V_\psi = V_\phi L,$$

then L and every U_i are Clifford. In particular, no code in this family admits a transversal non-Clifford logical unitary.

Rains recovered the three Pauli axes from a diagonal correlation tensor, and Van den Nest, Dehaene, and De Moor used that mechanism in a minimum-support LU-to-LC criterion for qubit stabilizer states [Rai99, Theorem 13] [VdNDDM05, Theorem 1 and Corollary 1]. For qubits, Theorem 1.1 follows directly: Proposition 3.4 supplies all three Pauli axes on every minimum support. Englbrecht and Kraus classify all qubit stabilizer local symmetries [EK20], while graphical and algorithmic descriptions of qubit stabilizer LU equivalence are now available [CP24, BdJV24]. Here the AME support squeeze gives a short additive-Weyl proof at every prime power and drives the atlas and quantitative results.

Tan proves the general AME-QMDS LU-orbit correspondence and the symmetry-transversal dictionary [Tan26, Theorems 3.5 and 3.6], and computes the canonical four-qutrit symmetry group [Tan26, Theorem 5.3]. Corollary 1.4 instead records factorwise Cliffordness for conversions between two stabilizer-QMDS encoders; no new qualitative hierarchy obstruction is claimed. Perfect-tensor operator pushing and the encoder interpretation originate in holographic codes [PYHP15, Eq. (2.3) and Section 5.7]; the atlas makes those maps a complete invariant here. The Bell-pair boundary is sharp because $U \otimes \bar{U}$ supplies non-Clifford symmetries.

The proof is organized around the two causal chains just described. Section 2 fixes the additive stabilizer conventions and derives the AME support formula. Sections 3 and 4 prove exact rigidity, the atlas classification, and the encoder corollary. Section 5 proves cleaning-based rounding, robust atlas compatibility, and the affine-character obstruction. Section 6

states the trust boundary, and Section 7 records the scaling consequences. The appendices contain partial-Weyl recognition, detailed two- and k -uniform stability, and the weaker single-marginal and aggregate rounding mechanisms.

2 Stabilizer AME support geometry

Let $q = p^e$ and fix the additive character $\chi(a) = \exp(2\pi i \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a)/p)$. On the standard basis of $\mathbb{C}^q$, indexed by $\mathbb{F}_q$, put

$$X(a)|x\rangle = |x+a\rangle, \quad Z(b)|x\rangle = \chi(bx)|x\rangle.$$

For $v = (a, b) \in \mathbb{F}_q^2$, write $W_v = X(a)Z(b)$, with any fixed consistent phase convention when only its projective axis is used. The q^2 single-party operators W_v form an orthogonal basis of the single-party Hilbert–Schmidt space. A unitary is Clifford when conjugation permutes their projective axes $\mathbb{C}W_v$.

After choosing dual $\mathbb{F}_p$ -bases of $\mathbb{F}_q$, this Weyl system is the Pauli system of e p -level factors. Its projective Clifford normalizer has the standard exact sequence

$$1 \longrightarrow \mathbb{F}_p^{2e} \longrightarrow \operatorname{PCliff}(q) \longrightarrow \operatorname{Sp}_{2e}(\mathbb{F}_p) \longrightarrow 1, \quad (2.0)$$

including $p = 2$: the kernel is the projective Pauli group, and every trace-symplectic $\mathbb{F}_p$ -linear label map has a Clifford lift [DDM03, HDDM05].

For n parties, a pure stabilizer state is fixed by an abelian Pauli group $\mathcal{S}$ of order q^n . Its projective label group $L \leq (\mathbb{F}_q^2)^n$, viewed additively over $\mathbb{F}_p$, is maximal isotropic for the trace-symplectic form. This is the standard arbitrary-additive prime-power stabilizer correspondence [KKKS06, Theorems 13 and 15]. For each $\ell \in L$, write the stabilizer lift as $\widetilde{W}_\ell = \alpha_\ell W_\ell$, where $\alpha_\ell \neq 0$. Then

$$|\psi\rangle\langle\psi| = q^{-n} \sum_{\ell \in L} \widetilde{W}_\ell. \quad (2.1)$$

If A is a party set, put

$$L(A) = \{\ell \in L : \operatorname{supp}(\ell) \subseteq A\}.$$

Weyl orthogonality and partial trace give

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L(A)} \widetilde{W}_\ell. \quad (2.2)$$

In particular, ρ_A^ψ is maximally mixed exactly when $L(A) = \{0\}$. Thus a stabilizer state on $2m$ parties is AME exactly when it has no nonidentity stabilizer label supported on at most m parties.

Equations (2.1)–(2.2) are the only state-specific input to the exact rigidity theorem. The same support geometry also identifies any chosen party as the logical leg of a one-qudit stabilizer encoder on the remaining $2m - 1$ parties. The resulting code has parameters $[[2m - 1, 1, m]]_q$: erasure of at most $m - 1$ physical parties is correctable because the complementary reductions of the AME state are maximally mixed. We use this encoder interpretation twice, first for the exact transversal consequence and then for quantitative cleaning.

3 Full-Weyl marginals and LU rigidity

We first isolate the linear-algebra statement that recovers a diagonal tensor’s coordinate axes. It is the full-Weyl counterpart of the three-Pauli argument in [Rai99, VdNDDM05]. The

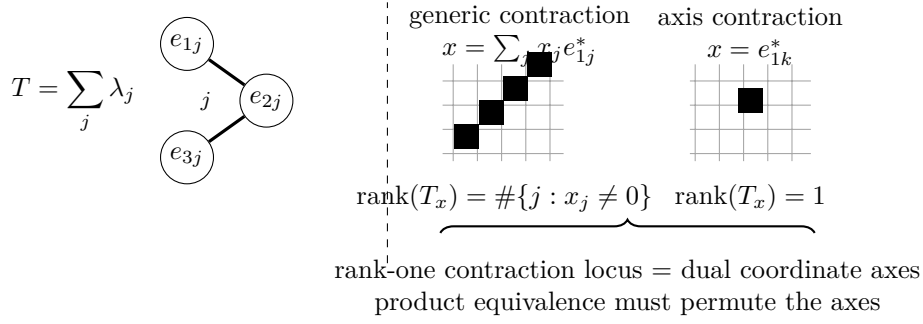


Figure 1: A diagonal tensor with all coefficients nonzero exposes its own axes. Contracting one leg against x and flattening the remainder gives rank equal to the number of nonzero coordinates of x , so the rank-one contraction locus is exactly the set of dual coordinate axes. For an $(m + 1)$ -party AME marginal in local Hilbert–Schmidt spaces, the recovered axes are the local Weyl axes, and a product conjugation must permute them.

statement itself is classical: it is Jennrich’s uniqueness theorem for trilinear decompositions, published in [Har70, §V] and there credited to Jennrich, whose conclusion that the factor matrices agree up to one permutation and compensating diagonal scalings is the monomial property below. It is the full-column-rank case of Kruskal’s condition [Kru77], extended to arbitrarily many factors in [SB00]; see [KB09, §3.2] for the attribution and for a survey of the uniqueness theory. We give a short self-contained proof because the contraction argument is the one used in Proposition 3.2, and claim no novelty for either the statement or the proof.

Lemma 3.1 (Diagonal-tensor axes). Let $r \geq 3$, let $E_1, \dots, E_r$ be N -dimensional complex inner product spaces, and choose orthonormal bases $\{e_{ij} : 1 \leq j \leq N\}$. If

$$T = \sum_{j=1}^N \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \quad \lambda_j \neq 0, \quad (3.1)$$

then the coordinate axes in every E_i are intrinsic to T . Consequently, a product of invertible linear maps carrying one tensor of the form (3.1) to another is monomial in the displayed bases.

Proof. For nonzero $x = \sum_j x_j e_{1j}^*$, contract the first factor against x . The result is

$$T_x = \sum_j \lambda_j x_j e_{2j} \otimes \cdots \otimes e_{rj}.$$

Flattening from E_2 to $E_3 \otimes \cdots \otimes E_r$ gives a matrix of rank $\#\{j : x_j \neq 0\}$. Thus T_x has rank one exactly when the ray of x is a dual coordinate axis. Product equivalence preserves this projective rank-one contraction locus. The inner products identify the recovered dual axes with the coordinate axes in the first factor. Apply the same argument in every factor. $\square$

Call an operator on $r \geq 3$ qudits *full-Weyl diagonal* if there is a q^2 -element set P , a distinguished element $0 \in P$, and bijections $f_i : P \rightarrow \mathbb{F}_q^2$ with $f_i(0) = 0$, for which

$$R = \sum_{v \in P} \lambda_v \bigotimes_{i=1}^r W_{f_i(v)}, \quad \lambda_v \neq 0. \quad (3.2)$$

The coefficients need not be equal. What matters is that the same q^2 -element index set runs bijectively through the complete local Weyl basis at every party. Neither P nor the bijections need be $\mathbb{F}_q$ -linear.

Proposition 3.2 (Full-Weyl marginal criterion). Let R and R' be full-Weyl diagonal operators on the same $r \geq 3$ parties, possibly with different indexing sets, bijections, and coefficients. If

$$R' = (U_1 \otimes \cdots \otimes U_r) R (U_1 \otimes \cdots \otimes U_r)^\dagger,$$

then every U_i is Clifford.

Proof. View R and R' as tensors in the local Hilbert–Schmidt spaces. After replacing W_v by the orthonormal basis $q^{-1/2}W_v$ and absorbing the common normalization into the coefficients, both have the form in Lemma 3.1, with $N = q^2$. Local conjugation acts invertibly on each factor, so it must permute the Weyl axes. It fixes the identity operator and hence the identity axis. It therefore permutes the nonidentity projective Weyl operators, which is exactly the single-qudit Clifford condition. $\square$

Corollary 3.3 (Full-Weyl marginal-cover rigidity). Let $|\psi\rangle, |\phi\rangle$ be n -qudit states and suppose a product unitary carries $|\psi\rangle$ to $|\phi\rangle$ up to phase, any party permutation having first been absorbed by relabelling $|\phi\rangle$, so that the two states are indexed by the same parties. If every party belongs to a set S of at least three parties for which both reduced operators ρ_S^ψ and ρ_S^ϕ are full-Weyl diagonal, then every local factor is Clifford.

Proof. Partial trace gives a product-unitary equivalence of the two reduced operators on each chosen S . Apply Proposition 3.2; the chosen sets cover all parties. $\square$

Proposition 3.4 (Stabilizer-AME support theorem). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state, and let A be any $m + 1$ parties. Then

$$|L(A)| = q^2.$$

Every nonidentity label in $L(A)$ has support exactly A , and for each $i \in A$ the coordinate projection

$$p_i : L(A) \longrightarrow \mathbb{F}_q^2$$

is bijective.

Proof. The projective label group L has order q^{2m} . Restrict its labels to the $m - 1$ parties outside A . The target has order $q^{2(m-1)}$, and the kernel is $L(A)$; hence

$$|L(A)| \geq q^{2m}/q^{2(m-1)} = q^2. \quad (3.3)$$

Fix $i \in A$. A label in the kernel of p_i is supported on the m -set $A \setminus \{i\}$. The AME condition and (2.2) exclude every such nonidentity stabilizer, so p_i is injective. Its codomain has order q^2 , giving the reverse inequality in (3.3) and hence bijectivity. The same no-small-support argument shows that every nonidentity element of $L(A)$ has full support A . $\square$

Proposition 3.5 (A stabilizer-AME marginal fixes the local Weyl axes). Let $|\psi\rangle, |\phi\rangle$ be stabilizer AME($2m, q$) states. If a product unitary on an $(m + 1)$ -set A carries ρ_A^ψ to ρ_A^ϕ , then every local factor on A is Clifford.

Proof. By (2.2),

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L_\psi(A)} \widetilde{W}_\ell.$$

Index this sum by $P = L_\psi(A)$. Proposition 3.4 says that every coordinate projection $P \rightarrow \mathbb{F}_q^2$ is bijective. Relative to the fixed Weyl convention, each stabilizer lift contributes a nonzero coefficient. Thus ρ_A^ψ is full-Weyl diagonal in the sense of (3.2), and the same holds for ρ_A^ϕ . Apply Proposition 3.2. $\square$

Proof of Theorem 1.1. Absorb the party permutation by relabelling $|\phi\rangle$. Partial trace gives a product-unitary equivalence $\rho_A^\psi \sim \rho_A^\phi$ for every $(m+1)$ -set A . Given a party i , choose such a set containing it and apply Proposition 3.5. Thus every U_i is Clifford. $\square$

4 Minimum-support atlases and encoder consequences

The same support calculation recovers standard quantum-MDS structure. If $|A| \leq m$, the AME condition gives $L(A) = \{0\}$. If $|A| > m$, then $\rho_{A^c}^\psi$ is maximally mixed, so purity of the global state gives $\text{Tr}((\rho_A^\psi)^2) = q^{-(2m-|A|)}$. On the other hand, (2.2) and Weyl orthogonality give $\text{Tr}((\rho_A^\psi)^2) = q^{-|A|}|L(A)|$. Hence, for every party set A ,

$$|L(A)| = q^{2 \max(|A|-m, 0)}. \quad (4.1)$$

Möbius inversion is the stabilizer-state specialization of the quantum-MDS Shor–Laflamme distribution of Huber and Grassl [HG20, Theorems 8 and 9]; in particular the number of weight- $m+1$ projective stabilizers is $\binom{2m}{m+1}(q^2-1)$. We use one further consequence. If $|A| \geq m+2$ and $i \neq j$ lie in A , then the dimensions in (4.1), together with

$$L(A \setminus \{i\}) \cap L(A \setminus \{j\}) = L(A \setminus \{i, j\}),$$

give

$$L(A) = L(A \setminus \{i\}) + L(A \setminus \{j\}). \quad (4.2)$$

Descending induction shows that the $(m+1)$ -supported subgroups span the full label group. This is the additive quantum-MDS form of minimum-weight generation, not a separate weight-enumerator claim.

We now prove the atlas classification stated in the introduction. Regard each local label group

$$P_i = \mathbb{F}_q^2$$

as a $2e$ -dimensional trace-symplectic space over $\mathbb{F}_p$. For every $(m+1)$ -set A and $i, j \in A$, the support theorem defines the operator-pushing transition

$$M_A(i, j) = p_j p_i^{-1} : P_i \longrightarrow P_j.$$

For two states ψ, ϕ on the same labelled parties, a *symplectic atlas equivalence* is a tuple $F_i \in \text{Sp}(P_i)$ satisfying

$$F_j M_A^\psi(i, j) = M_A^\phi(i, j) F_i \quad \text{for every } A \text{ and } i, j \in A. \quad (4.3)$$

Here $\text{Sp}(P_i) = \text{Sp}_{2e}(\mathbb{F}_p)$ is the action induced by the standard additive local Clifford group.

For one state, define

$$\begin{aligned} \Gamma_\psi &= \{[U] \in \text{PU}(q^{2m}) : U = U_1 \otimes \cdots \otimes U_{2m}, U|\psi\rangle \in \mathbb{C}|\psi\rangle\}, \\ \mathcal{G}_\psi &= \{(F_i) \in \prod_i \text{Sp}(P_i) : F_j M_A(i, j) = M_A(i, j) F_i \text{ for all } A, i, j \in A\}. \end{aligned}$$

Theorem 1.1 and the Clifford sequence (2.0) give a well-defined homomorphism

$$\lambda_\psi : \Gamma_\psi \longrightarrow \mathcal{G}_\psi, \quad [U_1 \otimes \cdots \otimes U_{2m}] \longmapsto (\overline{U}_i)_i,$$

where $\overline{U}_i$ is the induced symplectic label map. A product decomposition is unique up to one-site scalars, which do not change $\overline{U}_i$. The injection $L_\psi \hookrightarrow \Gamma_\psi$ sends a label ℓ to the projective Pauli $[W_\ell]$.

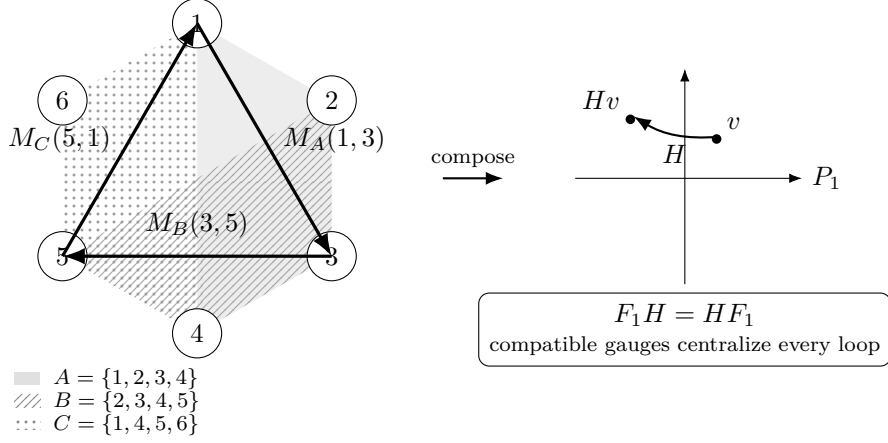


Figure 2: Operator pushing on minimum supports, shown for six parties. Each $(m + 1)$ -party support identifies the local Weyl-label planes of its parties through the bijective projections of the support theorem; overlapping supports compose to a loop whose holonomy $H = M_C M_B M_A$ is a linear map of the base plane, carrying a label v of the base plane to Hv . An atlas gauge is compatible precisely when it intertwines every transition; for one state over a prime field, its base block centralizes every loop holonomy.

Lemma 4.1 (Pauli phase correction). Let ψ and ϕ be stabilizer states with Lagrangian label groups L_ψ and L_ϕ . If a product Clifford U induces a symplectic label map carrying L_ψ onto L_ϕ , then there is a product Pauli W_v such that $W_v U|\psi\rangle \in \mathbb{C}|\phi\rangle$.

Proof. The state $U|\psi\rangle$ is stabilized on the same Pauli axes as $|\phi\rangle$, possibly with a different stabilizer character. The trace-symplectic pairing identifies the ambient label space modulo $L_\phi = L_\phi^\perp$ with the character dual of L_ϕ . Choose v whose pairing character is the inverse of the discrepancy. Conjugation by the product Pauli W_v cancels it. The corrected state and $|\phi\rangle$ therefore lie in the same one-dimensional joint stabilizer eigenspace. $\square$

Proof of Theorem 1.3. First fix the party labels. A block symplectic map $F = \bigoplus_i F_i$ carrying L_ψ to L_ϕ carries every supported subgroup $L_\psi(A)$ to $L_\phi(A)$. Writing these subgroups as graphs through any projection p_i gives exactly (4.3).

Conversely, suppose (4.3) holds. Fix A and $i \in A$. Every element of $L_\psi(A)$ is uniquely determined by its i -coordinate, and the equations with $j \in A$ show that its image under F is the unique element of $L_\phi(A)$ with i -coordinate $F_i p_i(\ell)$. Thus $F L_\psi(A) = L_\phi(A)$. Since the $(m + 1)$ -supported subgroups generate the full label groups, $F L_\psi = L_\phi$.

Sequence (2.0) supplies local Clifford lifts of the F_i , and Lemma 4.1 supplies the product-Pauli correction. Hence an atlas equivalence lifts to an LC equivalence. The reverse implication follows from the induced label action, and Theorem 1.1 replaces LC by LU. A party permutation is absorbed by relabelling first.

For automorphisms, λ_ψ is surjective by the same lifting and phase-correction argument. Its kernel consists of projective product Paulis by (2.0). Such a Pauli fixes the state ray exactly when its label lies in $L_\psi^\perp = L_\psi$. Thus the displayed injection of L_ψ identifies it with $\ker \lambda_\psi$, proving exactness.

Finally suppose $q = p$. Form the directed multigraph with vertices the parties and an edge (A, i, j) carrying $M_A(i, j)$. Every pair of parties belongs to an $(m + 1)$ -set, so the graph is connected. Choose a base b , a spanning tree, and tree transports $T_i : P_b \rightarrow P_i$, with $T_b = I$. For every edge define its based holonomy

$$H_A(i, j) = T_j^{-1} M_A(i, j) T_i \in \text{GL}_2(q).$$

Compatibility propagates a base block F_b uniquely as $F_i = T_i F_b T_i^{-1}$; the edge (A, i, j) is then compatible exactly when F_b commutes with $H_A(i, j)$. Since $\mathrm{Sp}_2(\mathbb{F}_q) = \mathrm{SL}_2(q)$ is normal in $\mathrm{GL}_2(q)$, every propagated block remains symplectic. Evaluation at the base therefore identifies $\mathcal{G}_\psi$ with the holonomy centralizer inside $\mathrm{SL}_2(q)$. Different spanning trees generate the same based holonomy subgroup; changing the base conjugates that subgroup, and hence its centralizer, by a chosen transport. $\square$

4.1 Encoder consequences

Proof of Corollary 1.4. The input marginal of each AME Choi state is $q^{-1}I$, so one-leg reshaping gives an isometry. The projection of its stabilizer label group onto the input Weyl-label space is surjective: otherwise a nonzero local label orthogonal to the image would lie in the maximal isotropic space $L^\perp = L$, contradicting the absence of a one-party stabilizer. The kernel therefore has q^{2m-2} elements and stabilizes the image, making it a one-logical-qudit stabilizer code. AME corrects erasure of any $m - 1$ outputs, so its distance is at least m ; the quantum Singleton bound gives equality. This is the standard perfect-tensor/quantum-MDS correspondence used in [Tan26, Theorems 3.5 and 3.6].

Let $|\Phi\rangle = q^{-1/2} \sum_x |x, x\rangle$, and use the Choi convention $|\psi\rangle = (I \otimes V_\psi)|\Phi\rangle$ and $|\phi\rangle = (I \otimes V_\phi)|\Phi\rangle$. Writing $U_{\mathrm{phys}} = \bigotimes_i U_i$, the relation $U_{\mathrm{phys}} V_\psi = V_\phi L$ gives

$$(I \otimes U_{\mathrm{phys}})|\psi\rangle = (L^T \otimes I)|\phi\rangle.$$

Hence

$$((L^T)^{-1} \otimes U_{\mathrm{phys}})|\psi\rangle = |\phi\rangle.$$

Theorem 1.1 makes every displayed local factor Clifford. Entrywise conjugation preserves the finite-field Weyl group, since $\overline{X(a)} = X(a)$ and $\overline{Z(b)} = Z(-b)$; it therefore preserves the Clifford normalizer. Taking adjoints preserves the normalizer as well, and $U^T = (\overline{U})^\dagger$. Thus transposition preserves the finite-field Clifford group, and L is Clifford. $\square$

5 Cleaning-based quantitative rounding

There is a stronger global route which does not estimate support sectors at all. Choose one AME party as a logical input. The other $2m - 1$ parties form an $[[2m - 1, 1, m]]_q$ stabilizer code, and the product of their local factors approximately implements the transpose-inverse of the factor on the chosen party. Three correctable regions then test that logical action by a nested commutator. The exact mechanism is standard: the stabilizer cleaning lemma is due to Bravyi and Terhal [BT09, Lemma 1], nested-commutator localization underlies the Bravyi–König hierarchy theorem [BK13], and Pastawski–Yoshida prove directly that a transversal logical unitary on three cleanable regions is Clifford [PY15, Lemma 5]. Disjointness gives a complementary exact-code hierarchy bound [JOKY18, Theorem 5]. The point below is therefore not a new qualitative transversal-gate obstruction. It is the robust step: an implementation with leakage error ε forces every nested logical Weyl commutator within 8ε of a scalar, after which a Fourier argument gives an explicit Clifford distance.

For an encoding isometry $V : \mathbb{C}^q \rightarrow \mathcal{H}$, write

$$E(T, L) = q^{-1/2} \|TV - VL\|_{\mathrm{HS}}.$$

It satisfies $E(TS, LM) \leq E(T, L) + E(S, M)$ and $E(T^\dagger, L^\dagger) = E(T, L)$.

Lemma 5.1 (Quantitative cleaning commutator). Let V encode one logical qudit in a stabilizer code whose physical parties have a partition into three correctable regions. Let T be

transversal, and suppose $E(T, L) \leq \varepsilon$. For all logical Weyl operators P, Q ,

$$q^{-1/2} \min_{|z|=1} \|[[L, P], Q] - zI\|_{\text{HS}} \leq 8\varepsilon, \quad (5.1)$$

where $[A, B] = ABA^\dagger B^\dagger$.

Proof. Let R_1, R_2, R_3 be the assumed correctable partition. The stabilizer cleaning lemma supplies a representative $\tilde{P}$ of P outside R_1 and a representative $\tilde{Q}$ of Q outside R_2 . For reference, cleaning here is the elementary symplectic statement that the restriction of a normalizer label to a correctable region lies in the restriction of the stabilizer label space; subtracting that stabilizer removes the restriction.

Transversality makes $[T, \tilde{P}]$ supported outside R_1 , so $C = [[T, \tilde{P}], \tilde{Q}]$ is supported on $R_1^c \cap R_2^c = R_3$. The composition rules give

$$E(C, [[L, P], Q]) \leq 4\varepsilon. \quad (5.2)$$

Knill–Laflamme correction on R_3 gives $V^\dagger C V = cI$. Hence the logical nested commutator lies within 4ε of cI . Its normalized Hilbert–Schmidt norm is one, so $|c| \geq 1 - 4\varepsilon$; replacing c by its phase proves (5.1). Notice that C need not preserve the code. Only its scalar compression is used, while (5.2) controls leakage. $\square$

The remaining step is finite-dimensional Fourier concentration. It is stated separately because it is the point where scalar commutators become Clifford axes.

Lemma 5.2 (Nested Weyl commutators round to a Clifford). Let $L \in U(q)$, $q = p^e$, and suppose for all Weyl labels v, w that

$$q^{-1/2} \min_{|z|=1} \|[[L, W_v], W_w] - zI\|_{\text{HS}} \leq \eta.$$

If

$$\eta < \tau_p := \min \left\{ \frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}} \right\},$$

then some additive one-qudit Clifford K satisfies

$$q^{-1/2} \min_{|z|=1} \|L - zK\|_{\text{HS}} \leq \eta.$$

Proof. Put $A_v = L W_v L^\dagger$. Conjugation of W_w by W_v changes only its phase, so every $[A_v, W_w]$ is within η of a scalar. We use the following Fourier observation twice. If $A = \sum_a c_a W_a$ is unitary and every $[A, W_w]$ is within η of a scalar, then A lies within η of one Weyl axis. Indeed, with $p_a = |c_a|^2$, the commutator estimate gives

$$\left| \sum_a p_a \chi_w(a) \right| \geq 1 - \eta^2/2.$$

Character orthogonality, averaged over w , yields $\sum_a p_a^2 \geq (1 - \eta^2/2)^2$. Thus $\max_a p_a \geq (1 - \eta^2/2)^2$, which is precisely the asserted normalized distance bound.

Apply the observation to each A_v , and call its unique nearby Weyl label $F(v)$. The exact identity $A_{v+u} \in \mathbb{C} A_v A_u$ puts the axes labelled by $F(v+u)$ and $F(v)+F(u)$ within 3η . Distinct Weyl axes have phase-optimized distance $\sqrt{2}$, so $\eta < 1/3$ makes F additive. Comparing the exact commutator phases of A_v, A_u with those of their nearby Weyls costs at most 4η . The least nontrivial p -th-root chord is $2 \sin(\pi/p)$, so the stated bound makes F trace-symplectic.

Choose an additive Clifford K_0 inducing F . Then $K_0^\dagger L$ approximately fixes every Weyl axis, so all its Weyl commutators are within η of scalars. The Fourier observation puts it within η of a Weyl; multiplying that Weyl into K_0 gives the required Clifford. $\square$

At $\varepsilon = 0$, the preceding two lemmas recover the three-cleanable-region Clifford conclusion of [PY15, Lemma 5]. Their quantitative combination, and the constants in (5.1), are what will be used here.

Theorem 5.3 (Cleaning-based global rounding). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state, $q = p^e$, $m \geq 2$, and $n = 2m$. Put

$$d_p = (2 - 2p^{-1/2})^{1/2},$$

and

$$R_{n,q}^{\text{clean}} = \min \left\{ \frac{\tau_p}{8}, \frac{1}{4\sqrt{2q}}, \frac{1}{8\pi\sqrt{n}}, \frac{d_p}{1 + 4\pi\sqrt{n}} \right\}. \quad (5.3)$$

If $\varepsilon(U) < R_{n,q}^{\text{clean}}$, then, up to a global phase,

$$U = g \bigotimes_{i=1}^n e^{ih_i}, \quad g \in G(\psi),$$

where the h_i are traceless Hermitian, have spectral spread at most π , and satisfy

$$\left(\sum_i \|h_i\|_F^2 \right)^{1/2} \leq \pi\sqrt{q}\varepsilon(U). \quad (5.4)$$

Before the exact-symmetry step, the recovered local Cliffords obey

$$q^{-1/2} \min_{|z|=1} \|U_i - zK_i\|_{\text{HS}} \leq 8\varepsilon(U) \quad (5.5)$$

at every party. For fixed q , (5.3) has order $n^{-1/2}$, conditional on existence of the stated AME states.

Proof. Choose party i as the logical leg and write the AME Choi form as $q^{-1/2} \sum_a |a\rangle \otimes V_i |a\rangle$. The other local factors form a transversal physical unitary T_i , and vectorization gives a logical unitary L_i , obtained from U_i by transpose and inverse, with $E(T_i, L_i) = \varepsilon(U)$. Lemmas 5.1 and 5.2, with $\eta = 8\varepsilon(U)$, give (5.5); transpose and inverse preserve the additive Clifford group.

Put $K = \bigotimes_i K_i$, choose principal logarithms of $K_i^\dagger U_i$ after optimizing the local phases, and let $D^2 = \sum_i \|h_i\|_F^2$. The chord-angle bound gives

$$D^2 \leq \frac{\pi^2 q}{4} \sum_i \left(q^{-1/2} \|U_i - K_i\|_{\text{HS}} \right)^2 \leq 16\pi^2 q n \varepsilon(U)^2. \quad (5.6)$$

The second clause of (5.3) puts every spectrum in an arc of length π ; the third gives $D \leq \sqrt{q}/2$. Two-uniformity gives $\|(K^\dagger U - I)\psi\| \leq D/\sqrt{q} \leq 4\pi\sqrt{n}\varepsilon(U)$. The fourth clause and the triangle inequality put $K\psi$ less than d_p from ψ , so Corollary B.23 makes K an exact symmetry. Apply Theorem B.18 to the residual to obtain (5.4). $\square$

The theorem already contains a genuinely collective estimate once the exact branch has been selected. Writing $g = \bigotimes_i g_i$ in the factorization above,

$$\sum_i \left(q^{-1/2} \min_{|z|=1} \|U_i - zg_i\|_{\text{HS}} \right)^2 \leq \frac{D^2}{q} \leq \pi^2 \varepsilon(U)^2. \quad (5.7)$$

The first inequality is the elementary chord bound $\|e^{ih_i} - I\|_{\text{HS}} \leq \|h_i\|_F$. Thus the $\sqrt{n}$ loss is absent after exact-symmetry recovery.

This does not enlarge the entry radius in (5.3). Before the branch is known, cleaning supplies only the separate bounds (5.5) for locally chosen Cliffords K_i . The n encoded implementation errors are unitary reshaping of the same global defect vector, not projections onto orthogonal sectors, so their squared bounds cannot be summed by a Bessel argument. A radius-improving collective estimate would itself imply constant-radius branch selection: the chord-angle inequality would make $\sum_i \|h_i\|_F^2 = O(q\varepsilon^2)$, and the uniform stabilizer gap would force $\bigotimes_i K_i$ to be exact. The missing input is therefore compatibility of the recovered discrete Clifford data, not a sharper final norm summation.

The linear part of that compatibility can in fact be recovered without a party-count loss.

Proposition 5.4 (Robust symplectic-atlas compatibility). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state, $q = p^e$, and let $U = \bigotimes_i U_i$ be a product unitary. Suppose

$$\varepsilon(U) < R_q^{\text{atlas}} := \min \left\{ \frac{\tau_p}{8}, \frac{2 \sin(\pi/p)}{4 + 64\sqrt{q}} \right\}.$$

The cleaning and Fourier lemmas give local Cliffords K_i satisfying (5.5). Let F_i be their symplectic maps. Then

$$F_j M_A(i, j) = M_A(i, j) F_i$$

for every $(m+1)$ -set A and $i, j \in A$. Consequently $F = \bigoplus_i F_i$ preserves the stabilizer label group L , and some product-Pauli correction $W_v K$ is an exact symmetry of ψ .

Proof. Fix A , distinct $i, j \in A$, and put $B = A^c \cup \{i, j\}$. Both A and B have size $m+1$, while $A \cap B = \{i, j\}$. Choose $\ell \in L(A)$ and $r \in L(B)$. Using the stabilizer lifts $\widetilde{W}_\ell$ and $\widetilde{W}_r$ from (2.1), set

$$C = [U \widetilde{W}_\ell U^\dagger, \widetilde{W}_r] = [U W_\ell U^\dagger, W_r],$$

where the second equality holds because scalar lift phases cancel from a commutator. After choosing the phase of U realizing its defect, $\|(U \widetilde{W}_\ell U^\dagger - I)\psi\| \leq 2\varepsilon(U)$, and the same holds with the adjoint. Since $\widetilde{W}_r \psi = \psi$,

$$\|(C - I)\psi\| \leq 4\varepsilon(U). \quad (5.8)$$

The commutator is supported only on $A \cap B = \{i, j\}$. Choose phases so that $\|U_s - K_s\|_{\text{HS}} \leq 8\sqrt{q}\varepsilon(U)$. Replacing $U_s W_{\ell_s} U_s^\dagger$ by its K_s -conjugate costs at most $16\sqrt{q}\varepsilon(U)$ in operator norm, and replacing its commutator with W_{r_s} costs at most twice that. On the two sites, the total cost is therefore at most $64\sqrt{q}\varepsilon(U)$. The Clifford commutator is a scalar p -th root ζ , so (5.8) gives

$$|\zeta - 1| \leq (4 + 64\sqrt{q})\varepsilon(U) < 2 \sin(\pi/p).$$

Hence $\zeta = 1$.

Write $a = \ell_i$, $b = r_i$, $\ell_j = M_A(i, j)a$, and $r_j = M_B(i, j)b$. The identity $\zeta = 1$, for all a, b , says

$$[F_i a, b] + [F_j M_A(i, j)a, M_B(i, j)b] = 0. \quad (5.9)$$

Here brackets denote the trace-symplectic pairing. Isotropy of L , applied to the element of $L(A)$ whose i -coordinate is $F_i a$, gives

$$[F_i a, b] + [M_A(i, j)F_i a, M_B(i, j)b] = 0.$$

Subtracting this from (5.9), using surjectivity of $M_B(i, j)$ and nondegeneracy of the pairing, proves the atlas equation. The exact atlas theorem gives $FL = L$, and Lemma 4.1 supplies W_v . $\square$

The Pauli correction in the proposition is essential. All localized commutators are unchanged when a recovered K_i is multiplied by a local Weyl: that multiplication changes conjugation phases, and the commutator cancels them. Equivalently, the symplectic atlas sees L but not its stabilizer character. The correction v can therefore be nonzero modulo L , in which case $W_v K$ need not be locally close to U . Exact affine compatibility cannot be obtained from these commutator tests, and Proposition 5.4 alone does not improve (5.3). Controlling the character discrepancy is the same remaining branch-selection problem isolated by the collective estimate above.

The exact formula, rather than a fixed-dimension slogan, controls scaling. Uniformly over prime powers $q = p^e$ and existing tensors,

$$R_{n,q}^{\text{clean}} = \Theta\left(\min\{p^{-1}, q^{-1/2}, n^{-1/2}\}\right), \quad (5.10)$$

with absolute implied constants. Indeed $\sin(\pi/p) = \Theta(p^{-1})$, while d_p is bounded above and below by positive absolute constants. An asymptotic use of (5.10) must specify an existing family.

For the explicit generalized and extended generalized Reed–Solomon construction, $n = 2m \leq q + 1$, so the party-count term cannot be smaller than the local-dimension term. Hence

$$R_{n,q}^{\text{clean}} = \begin{cases} \Theta(q^{-1}), & q = p, \\ \Theta(q^{-1/2}), & q = p^e, e \geq 2. \end{cases} \quad (5.11)$$

On a length-saturating subfamily $n \asymp q$, these are $\Theta(n^{-1})$ and $\Theta(n^{-1/2})$, respectively. For fixed characteristic and growing extension degree, the second line applies. The earlier fixed- q , $n^{-1/2}$ reading of the exact minimum remains a conditional formal slice only; it is not an AME-existence claim.

If the sharper local constant and the old global generator budget are needed, the same proof gives the conclusion of Corollary B.19 under

$$\varepsilon(U) < \min\left\{\frac{\tau_p}{8}, \frac{1}{16\pi n\sqrt{q}}\right\}. \quad (5.12)$$

Indeed (5.5), the operator-norm bound by Hilbert–Schmidt norm, and the principal logarithm give $\sum_i \|h_i\|_{\text{op}} \leq 8\pi n\sqrt{q}\varepsilon(U) < 1/2$; the sharp local theorem then gives $D \leq \sqrt{6q/5}\varepsilon(U)$.

The cleaning route supersedes both explicit radii above on every admissible parameter pair where its displayed minimum is larger, and it has no boundary at $q = 4$. The earlier theorems remain useful because they expose the exact cost of reading one marginal and of aggregating orthogonal minimum-support sectors. The new $n^{-1/2}$ scale is a proved lower bound, not an optimality claim; no product-unitary family is known to saturate the collective frame-error estimate (5.6).

6 Verification and trust boundary

The principal results are conceptual proofs. This paper contains no paper-facing finite census, generated certificate, or numerical experiment. Its trust boundary has three layers: complete manuscript proofs, selected kernel-checked algebraic cores, and standard results imported by citation.

The formal source namespace is `RelativeConicArcs.AMELU`. In the checked Lean toolchain `v4.32.0-rc1`, the relevant exact cores lie in the support-profile, tensor-rigidity, holonomy-centralizer, stabilizer-dictionary, encoder-transversal, and multipartite modules. Their aggregate audit reports only `propext`, `Classical.choice`, and `Quot.sound` for these declarations.

Table 1: Trust boundary for the theorem families retained in this paper. “Core formalization” means that the named algebraic step is kernel checked; it does not promote the complete manuscript theorem to a formal theorem.

Result family	Manuscript proof	Formal status
AME support squeeze and full-Weyl rigidity	Sections 2–3	coordinate squeeze, unique local labels, and the abstract axis criterion are kernel checked; the concrete additive-state marginal composition is not
Minimum-support atlas and phase correction	Section 4	minimum-support generation, the abstract holonomy centralizer, and the stabilizer-character algebra are kernel checked; the state-ray correction is a manuscript step
Encoder Clifford no-go	Section 4	the Choi orientation and Clifford closure are kernel checked; the complete arbitrary-additive encoder theorem is assembled in the manuscript
Cleaning and Weyl–Fourier rounding	Section 5	manuscript only; neither the leakage constant 8, the Fourier rounding lemma, nor the global radius is formalized
Robust atlas compatibility and affine obstruction	Section 5	manuscript only
Two- and k -uniform stability and alternative radii	Appendices A–B	only the generator splitting, single-exponential identity, polarized second-moment identity, and absence of a nonscalar continuous symmetry are kernel checked; all quantitative constants are manuscript only

That aggregate spans a wider development than this paper: it also carries the six-party MDS–CSS applications that appear in the companion paper [Rud26], so it is not a closure specific to the results here, and no content-addressed contract identifies a subset of it as this paper’s own. Each row of Table 1 therefore states which algebraic core is kernel checked, and the paper claims no end-to-end formalization of any theorem family.

The cleaning theorem uses three cited exact inputs: stabilizer cleaning, localization of nested commutators, and the exact three-cleanable-region Clifford obstruction. The manuscript proves the leakage estimate, the finite Weyl–Fourier concentration step, the stabilizer overlap gap, and their AME composition. Equations (5.3)–(5.11) are closed-form deductions; they do not depend on a computational search or floating-point evaluation.

The appendices serve as mechanism comparisons. Their single-marginal and aggregate thresholds are weaker than the cleaning radius, and none is used to prove Theorem 1.2. The partial-Weyl recognition criterion is likewise independent of the stabilizer-AME support squeeze used in the headline theorem.

7 Conclusion

The same support geometry controls the exact and approximate problems. On an $(m + 1)$ -party marginal it exposes every local Weyl axis, which turns arbitrary product-unitary intertwiners into product Cliffords and organizes their residual freedom as a minimum-support atlas. After one party is reinterpreted as a logical input, the support geometry becomes three-region correctability. Cleaning and Weyl–Fourier concentration then recover the same discrete Clifford frame without paying the exponentially small amplitude of a single marginal.

The quantitative conclusion is uniform only when all varying parameters are displayed:

$$R_{n,q}^{\text{clean}} = \Theta\left(\min\{p^{-1}, q^{-1/2}, n^{-1/2}\}\right), \quad q = p^e.$$

For fixed local dimension this gives the conditional $n^{-1/2}$ scale along any existing stabilizer-AME family. On generalized and extended Reed–Solomon constructions, where $n \leq q + 1$, the scale is $\Theta(q^{-1})$ over prime fields and $\Theta(q^{-1/2})$ when $e \geq 2$. These are certified radii; their optimal order is open.

The robust atlas result identifies the remaining bottleneck. Linear symplectic compatibility becomes exact at a radius depending only on the local dimension, but commutators discard the affine Weyl phases. They therefore determine the stabilizer label group and not its character. A party-count-independent global entry theorem would need a new way to control that character discrepancy, or an equivalent collective branch selection principle, before the product-Pauli correction is known to be locally small.

A Partial-Weyl recognition beyond AME states

The full-Weyl criterion recovers the entire local Weyl basis from one marginal. More generally, several marginals may each recover only part of that basis, while their labels together generate it. The following criterion isolates this mechanism. It uses only a finite abelian label group and a Weyl basis whose multiplication is additive up to phase.

Call an operator on r parties *partial-Weyl diagonal* if there are a finite index set P with a distinguished element 0, injections $f_i : P \rightarrow \mathbb{F}_q^2$ with $f_i(0) = 0$, and nonzero coefficients, for which

$$R = \sum_{v \in P} \lambda_v \bigotimes_{i=1}^r W_{f_i(v)}, \quad \lambda_v \neq 0.$$

Write $N = |P|$. The full-Weyl condition (3.2) is the case $N = q^2$, where the injections are forced to be bijections.

Proposition A.1 (Partial-Weyl marginal criterion). Let R and R' be partial-Weyl diagonal operators on the same r parties with index sets of the same size $N \geq 2$, possibly with different index sets, injections, and coefficients, and suppose

$$R' = (U_1 \otimes \cdots \otimes U_r) R (U_1 \otimes \cdots \otimes U_r)^\dagger.$$

Assume $r \geq 3$, or $N = 2$. Then for each i , conjugation by U_i carries every $W_{f_i(v)}$ to a phase times a Weyl operator, and induces a bijection $f_i(P) \rightarrow f'_i(P')$ fixing 0.

Proof. Suppose $r \geq 3$. Let E_i be the span of $\{W_{f_i(v)} : v \in P\}$ inside the single-party Hilbert–Schmidt space and E'_i the span of $\{W_{f'_i(v')} : v' \in P'\}$. Both have dimension N , because the Weyl operators are linearly independent and the injections are injective, and the rescaled operators $q^{-1/2}W_v$ are orthonormal. All N coefficients of R are nonzero, so by the rank computation in the proof of Lemma 3.1 the smallest subspace of factor i supporting R is E_i ,

and likewise for R' and E'_i ; hence the product of the conjugations carries E_i onto E'_i . Now apply Lemma 3.1 to the two tensors in $E_1 \otimes \cdots \otimes E_r$ and $E'_1 \otimes \cdots \otimes E'_r$. Its proof uses only $r \geq 3$, nonzero coefficients, and orthonormal bases, never $N = q^2$. The maps are therefore monomial in the displayed bases, and each scalar is a phase because conjugation is unitary for the Hilbert–Schmidt inner product. The identity axis is fixed, so the bijection fixes 0.

Suppose instead $N = 2$, with $R = \lambda_0 I + \lambda_1 \bigotimes_i W_{f_i(1)}$ and $R' = \lambda'_0 I + \lambda'_1 \bigotimes_i W_{f'_i(1)}$. Conjugation fixes the identity, so comparing traces gives $\lambda_0 = \lambda'_0$; subtracting leaves an equality of nonzero product operators, whose tensor factorization gives $U_i W_{f_i(1)} U_i^\dagger = c_i W_{f'_i(1)}$ with $\prod_i c_i = \lambda'_1 / \lambda_1$, and each c_i is a phase by unitarity. $\square$

The arity condition cannot be removed in this form. For two parties and three or more terms, equal coefficient moduli leave a unitary freedom among the corresponding Schmidt axes. With two terms, conjugation fixes the identity term and hence singles out the other axis.

Fix a product unitary carrying $|\psi\rangle$ to $|\phi\rangle$ up to phase, after absorbing any party permutation by relabelling $|\phi\rangle$. For each party i put

$$\Lambda_i = \{a \in \mathbb{F}_q^2 : U_i W_a U_i^\dagger \in \mathbb{C}^\times W_b \text{ for some } b\},$$

the *recognition group* at i .

Lemma A.2 (Recognition group). Λ_i is a subgroup of $\mathbb{F}_q^2$, the induced map $a \mapsto b$ is an injective additive homomorphism, and U_i is Clifford if and only if $\Lambda_i = \mathbb{F}_q^2$.

Proof. If $U_i W_a U_i^\dagger = \alpha W_{a'}$ and $U_i W_b U_i^\dagger = \beta W_{b'}$, then $W_a W_b$ is a phase times W_{a+b} , so conjugating that product shows $U_i W_{a+b} U_i^\dagger$ is a scalar times $W_{a'+b'}$. Hence Λ_i is closed under addition and the induced map is additive; it is injective because distinct labels have distinct projective axes. If $\Lambda_i = \mathbb{F}_q^2$ the induced map is an injective endomorphism of a finite group, hence bijective, so conjugation permutes all projective Weyl axes, which is the Clifford condition. The converse is immediate. $\square$

When $q = 2$, a recognition group containing a nonzero label is either a line or all of $\mathbb{F}_2^2$. The criterion below assembles labels recovered from different marginals; in larger local dimension it also accommodates intermediate subgroups that have no qubit analogue.

Corollary A.3 (Generation criterion). Suppose that for each party i there is a family of party sets containing i , each with at least three parties or with a two-element index set, such that ρ_S^ψ and ρ_S^ϕ are partial-Weyl diagonal with index sets of equal size. If the labels recovered at i over that family generate $\mathbb{F}_q^2$, then U_i is Clifford.

Proof. Partial trace gives a product-unitary equivalence of the two reduced operators on each set. Proposition A.1 places each recovered label in Λ_i , and Lemma A.2 closes Λ_i under addition and converts $\Lambda_i = \mathbb{F}_q^2$ into the Clifford property. $\square$

Corollary 3.3 is the case of one set per party with $N = q^2$, where generation is automatic. For stabilizer states the supply of usable sets is exactly the minimal supports.

Lemma A.4 (Minimal supports carry partial-Weyl marginals). Let $|\psi\rangle$ be a stabilizer state and ω a minimal support of its stabilizer, and let $L(\omega)$ be the group of labels supported inside ω . Then for every $i \in \omega$ the coordinate projection $\pi_i : L(\omega) \rightarrow \mathbb{F}_q^2$ is injective. Hence $L(\omega)$ is isomorphic to a subgroup of $\mathbb{F}_q^2$, every nonzero element of $L(\omega)$ has support exactly ω , and ρ_ω is partial-Weyl diagonal with index set $L(\omega)$ and injections π_i . Conversely, if the retained label group on a party set has at least two elements and injects at every party, that set is a minimal support.

Proof. An element of the kernel of π_i has support contained in $\omega \setminus \{i\}$, a proper subset of ω , so it is zero by minimality. The partial trace of $|\psi\rangle\langle\psi|$ over the complement of ω retains exactly the stabilizer elements supported inside ω , each with a coefficient of modulus one. For the converse, injectivity at every party of the set says that no nonzero element of its label group vanishes anywhere on it, so no nonzero element has smaller support. $\square$

At $q = 2$ the possible orders of $L(\omega)$ are 2 and 4, which is the dichotomy of [VdNDDM05, Lemma 1]; the intermediate orders available for $q > 2$ are invisible there. They occur naturally in CSS states: a minimum-weight X -support carrying no Z -word contributes a line of q labels.

For two LU-equivalent stabilizer states, marginal purity preserves both the minimal supports and the order of each retained label group, since

$$\mathrm{Tr}(\rho_\omega^2) = \frac{|L(\omega)|}{q^{|\omega|}}.$$

Thus every source chart above has a target chart of the same size. In the qubit case, the hypothesis of [VdNDDM05, Theorem 1] says exactly that the labels from all minimal supports through a party generate $\mathbb{F}_2^2$. Label groups of order two use the two-term case of Proposition A.1; label groups of order four use its $r \geq 3$ case. Their standing assumptions of at least three parties and full entanglement exclude a two-party support in the latter case. Corollary A.3 therefore recovers their full theorem, not only its single-support corollary.

Corollary A.5 (CSS states over a prime field). Let q be prime and let $|\psi\rangle$ be a CSS coset state such that every coordinate lies in two minimal supports of the stabilizer, each with at least three coordinates, one whose label group contains a nonzero X -type label and one whose label group contains a nonzero Z -type label. Then every product-unitary equivalence to another such state has Clifford factors.

Proof. The preceding purity argument supplies matching partial-Weyl marginals for the target state. The two supports contribute independent labels $(x, 0)$ and $(0, z)$ to Λ_i , which generate $\mathbb{F}_q^2$ over a prime field. Apply Corollary A.3. $\square$

A minimum-weight codeword of the X code need not give a minimal support of the stabilizer, since a Z -word may sit strictly inside it; the hypothesis above is about minimal supports of the full stabilizer. What is not automatic for an arbitrary additive stabilizer is either coverage by minimal supports or the availability of both label types at every party; the corollary keeps both requirements explicit.

Remark A.6 (Arbitrary local dimension). Nothing in the partial-Weyl condition, Proposition A.1, Lemma A.2, Corollary A.3, or Lemma A.4 uses the field structure of $\mathbb{F}_q$. Each uses only that the labels form a finite abelian group whose Weyl operators are an orthogonal Hilbert–Schmidt basis multiplying additively up to phase. Replacing $\mathbb{F}_q^2$ by $\mathbb{Z}_d^2$, with X the shift and Z the clock operator on $\mathbb{C}^d$, all five statements hold verbatim at every local dimension d , prime power or not. At composite d , a unitary whose conjugation permutes the projective Weyl axes still normalizes the Pauli group itself and is Clifford in the standard sense. Write $UW_aU^\dagger = cW_{a'}$. For odd d , every Weyl operator satisfies $W_a^d = I$; taking d -th powers gives $c^d = 1$. For even d , every Weyl operator satisfies $W_a^{2d} = I$, and the standard Pauli convention includes the $2d$ -th roots of unity; taking $2d$ -th powers gives $c^{2d} = 1$. Thus c is already a Pauli phase and $UW_aU^\dagger$ lies in the Pauli group. When d is a proper prime power, the $\mathbb{Z}_d$ Weyl system and the $\mathbb{F}_d$ Weyl system on $\mathbb{C}^d$ are different, and a criterion instantiated in one says nothing in the other.

The generation hypothesis is sufficient, not necessary. Wong and Jiang exhibit at $d = 9$ a three-party stabilizer state that is LU but not LC equivalent to the GHZ state [WJ25]. Its

three two-party minimal supports each carry nine labels whose projections lie in $3\mathbb{Z}_9 \times 3\mathbb{Z}_9$. They generate a subgroup of index nine rather than $\mathbb{Z}_9^2$, so Corollary A.3 does not apply.

Chang and Jing introduced the general Kruskal-type uniqueness method for coefficient tensors in a discrete operator basis into the qubit LU problem [CJ22]. Their generic Pauli tensors are not diagonal in that basis, and uniqueness yields invariants separating LU classes rather than a Clifford conclusion. For qubit stabilizer states that conclusion is [VdNDDM05, Theorem 1].

B Secondary quantitative routes and local stability

The rigidity theorem uses the stabilizer hypothesis to identify the symmetry group. Finiteness of that group needs far less: two-party marginals alone, at any local dimension, with no stabilizer structure. Throughout this subsection $q \geq 2$ is an arbitrary integer; only the statements naming Clifford operators return to $q = p^e$.

Call a pure state $|\psi\rangle \in (\mathbb{C}^q)^{\otimes n}$ *2-uniform* if

$$\rho_{\{j,k\}}^\psi = q^{-2}I \quad \text{for all } j \neq k.$$

Tracing once more makes every one-party marginal $q^{-1}I$. Such states exist only for $n \geq 4$, since at $n = 3$ purity forces $\text{rank } \rho_{\{1,2\}} = \text{rank } \rho_{\{3\}} \leq q$; and every $\text{AME}(2m, q)$ state with $m \geq 2$ is 2-uniform, stabilizer or not. Write

$$G(\psi) = \{U_1 \otimes \cdots \otimes U_n : (U_1 \otimes \cdots \otimes U_n)|\psi\rangle \in \mathbb{C}|\psi\rangle\}$$

for its product-unitary symmetry group and, for a product unitary U ,

$$\varepsilon(U) = \min_{\theta} \|U|\psi\rangle - e^{i\theta}|\psi\rangle\| = (2 - 2|\langle\psi|U|\psi\rangle|)^{1/2}$$

for its defect. Because operators on distinct sites commute, a product unitary is a single exponential: $\bigotimes_j e^{iH_j} = e^{iL}$ with $L = \sum_j H_j^{(j)}$, where $H^{(j)}$ denotes H acting at site j . Splitting each H_j into a traceless part h_j and a multiple of the identity leaves $L = M + cI$ with $M = \sum_j h_j^{(j)}$. Everything below rests on one computation about M .

Figure 3 gives a roadmap for the local estimates, the Clifford separation step, and the resulting explicit threshold.

Lemma B.1 (Local-generator isometry). Let $|\psi\rangle$ be 2-uniform and let h_j and h'_j , $1 \leq j \leq n$, be traceless Hermitian operators on $\mathbb{C}^q$. Put $M = \sum_j h_j^{(j)}$ and $M' = \sum_j h'_j{}^{(j)}$. Then $\langle\psi|M|\psi\rangle = 0$ and

$$\langle M|\psi\rangle, M'|\psi\rangle\rangle = \frac{1}{q} \sum_{j=1}^n \text{Tr}(h_j h'_j). \quad (\text{B.1})$$

In particular $\|M|\psi\rangle\|^2 = q^{-1} \sum_j \|h_j\|_F^2$, so $(h_1, \dots, h_n) \mapsto \sqrt{q} M|\psi\rangle$ is an isometry from the traceless local generators, carrying the product Frobenius form, into the Hilbert space regarded as a real inner product space.

Proof. The one-party marginals give $\langle M \rangle = \sum_j \text{Tr}(\rho_{\{j\}}^\psi h_j) = \sum_j \text{Tr}(h_j)/q = 0$. Expand $\langle\psi|MM'|\psi\rangle$. The terms with equal site indices contribute $\text{Tr}(\rho_{\{j\}}^\psi h_j h'_j) = \text{Tr}(h_j h'_j)/q$. A term with $j \neq k$ contributes $\text{Tr}(\rho_{\{j,k\}}^\psi (h_j \otimes h'_k)) = \text{Tr}(h_j) \text{Tr}(h'_k)/q^2 = 0$; both maximal mixedness of the pair marginal and tracelessness are used. Hence $\langle\psi|MM'|\psi\rangle = q^{-1} \sum_j \text{Tr}(h_j h'_j)$, which is real. Since M and M' are Hermitian, the imaginary part of $\langle M\psi, M'\psi \rangle$ is $\langle [M, M'] \rangle / 2i$, and it vanishes as well: operators at distinct sites commute, and the surviving one-site commutators are traceless, so each term is of the form $\text{Tr}(\rho_{\{j\}}^\psi [h_j, h'_j]) = \text{Tr}([h_j, h'_j])/q = 0$. A real symmetric bilinear form agreeing with the product Frobenius form is that form. $\square$

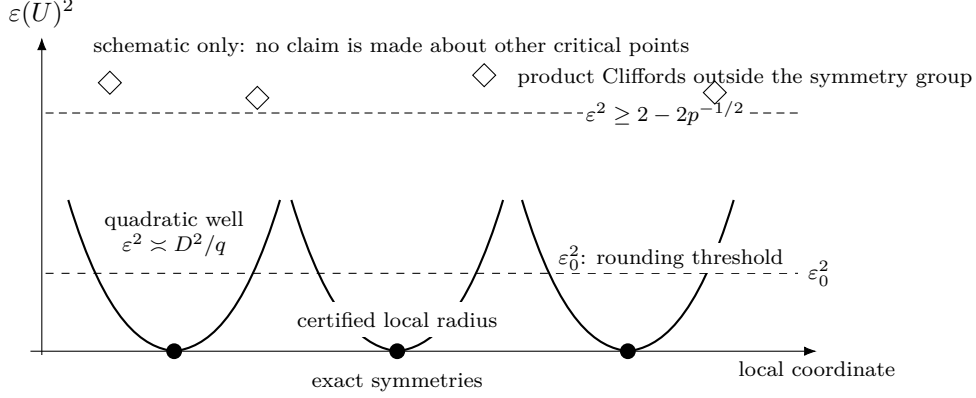


Figure 3: Roadmap for the defect estimates (a one-dimensional schematic of a high-dimensional product-unitary group). Exact symmetries are isolated zeros with certified local quadratic growth. Every product Clifford outside the symmetry group has defect at least $(2 - 2p^{-1/2})^{1/2}$; below the explicit decomposition threshold, an arbitrary product unitary rounds to a Clifford and then to an exact symmetry. The picture asserts neither that every local minimum is Clifford nor that the displayed coordinate exhausts the group.

The identity has no error term, and that is the content of the 2-uniformity hypothesis rather than a coincidence: the cross terms cancel exactly because the pair marginals are exactly $q^{-2}I$. A Bell pair is the boundary case. There $\rho_{\{1,2\}}$ is a pure state, the cross term survives, and $M = h \otimes I - I \otimes h^T$ annihilates the state for every h — the $U \otimes \bar{U}$ gauge line that makes $m \geq 2$ sharp in Theorem 1.1.

The second ingredient is that a one-parameter group of product unitaries has a generator of summed local form. This is not automatic: a one-parameter subgroup of $G(\psi)$ is a priori only a continuous homomorphism into $U(q^n)$, and the factors of a product unitary are determined only up to phases, so differentiating factor by factor carries a lifting obligation.

Lemma B.2 (Product generators). Let $\pi : U(q)^n \rightarrow U(q^n)$ send $(U_1, \dots, U_n)$ to $U_1 \otimes \dots \otimes U_n$. Then π is a smooth homomorphism of compact Lie groups, its image P is a closed subgroup of $U(q^n)$, and

$$\text{Lie}(P) = \left\{ i \sum_j H_j^{(j)} : H_j = H_j^\dagger \text{ on } \mathbb{C}^q \right\}. \quad (\text{B.2})$$

Hence every one-parameter subgroup of a closed subgroup of P has a generator of that form.

Proof. The tensor product of unitaries is multiplicative factor by factor, so π is a smooth homomorphism. Its source is compact, so P is compact and therefore closed, and Cartan's theorem makes P an embedded Lie subgroup of $U(q^n)$. A surjective homomorphism of Lie groups has surjective differential, so $\text{Lie}(P) = d\pi(\mathfrak{u}(q)^n)$, which is the displayed space; its kernel is $\{(i\tau_1 I, \dots, i\tau_n I) : \sum_j \tau_j = 0\}$. A closed subgroup $H \leq P$ is a Lie subgroup with $\text{Lie}(H) \subseteq \text{Lie}(P)$, and the generator of a one-parameter subgroup of H lies in $\text{Lie}(H)$. $\square$

Theorem B.3 (Discreteness from 2-uniformity). Let $|\psi\rangle$ be a 2-uniform pure state of n qudits of local dimension q . Then $\text{Lie}(G(\psi)) = i\mathbb{R}I$, and $G(\psi)/U(1) \cdot I$ is finite. In particular every $\text{AME}(2m, q)$ state with $m \geq 2$, stabilizer or not, has finite product-unitary symmetry group modulo global phase.

Proof. The set $\{U \in U(q^n) : U|\psi\rangle \in \mathbb{C}|\psi\rangle\}$ is a closed subgroup, so $G(\psi) = P \cap \{\dots\}$ is a closed subgroup of P and hence a compact Lie group. Let $X \in \text{Lie}(G(\psi))$. By Lemma B.2, $X = iL$ with $L = \sum_j H_j^{(j)}$ Hermitian; write $L = M + cI$ as above. Each e^{itL} fixes the ray of

$|\psi\rangle$, so $e^{itL}|\psi\rangle = \lambda(t)|\psi\rangle$ with $\lambda(t) = \langle\psi|e^{itL}|\psi\rangle$ unimodular and smooth in t . Differentiating at $t = 0$ shows that $|\psi\rangle$ is an eigenvector of the Hermitian operator L , say $L|\psi\rangle = \mu|\psi\rangle$ with μ real. Pairing with $|\psi\rangle$ and using $\langle M \rangle = 0$ gives $\mu = c$, hence $M|\psi\rangle = 0$. By Lemma B.1, $\sum_j \|h_j\|_F^2 = 0$, so every $h_j = 0$ and every H_j is scalar. Thus $\text{Lie}(G(\psi)) = i\mathbb{R}I$. Since $U(1) \cdot I$ is a closed central one-dimensional subgroup of $G(\psi)$, the compact Lie group $G(\psi)/U(1) \cdot I$ has dimension zero and is therefore finite. $\square$

Remark B.4 (Which torus is divided out). Upstairs in $U(q)^n$ the preimage $\pi^{-1}(G(\psi))$ always contains the local-phase torus $T = U(1)^n$, of dimension n . This does not contradict the theorem: $\pi(T) = U(1) \cdot I$ and $\ker \pi \subset T$ has dimension $n-1$, so $\pi^{-1}(G(\psi))/T \cong G(\psi)/U(1) \cdot I$. Statements about local factors below are statements on $\text{PU}(q)^n$, not statements modulo one global phase.

Remark B.5 (Discreteness in the literature). Finiteness of this kind is known in two special cases. Wirthmüller determines the identity component of the local-unitary automorphism group of a binary stabilizer code and gives a protected-qubit criterion for finiteness of the induced group [Wir11, Theorem 7 and Corollary 11]; his exclusion of Bell factors is the $n \geq 4$ boundary above. Englbrecht and Kraus determine the local symmetries of qubit stabilizer states, including where continuous ones occur [EK20]. Tan computes the local symmetry group of the four-qutrit AME state outright, of order 5832 [Tan26, Theorem B.3]. Theorem B.3 covers arbitrary local dimension and arbitrary, in particular nonstabilizer, 2-uniform states; to our knowledge no statement at that scope is available.

The hypothesis is exactly 2-uniformity, and one-party marginals are not enough. Słowiak, Sawicki, and Maciążek construct locally maximally entangled states — every one-party marginal maximally mixed — whose local-unitary symmetry group contains a nontrivial continuous image of $\text{SU}(N)$ and is therefore positive-dimensional, with dimension growing in the number of parties [SSM21, Lemma 2]. Passing from one-party to two-party marginals is what forces discreteness. In the language of geometric invariant theory the theorem says that a 2-uniform state is a stable point for $\text{SL}(q, \mathbb{C})^n$ rather than merely a critical one; stabilizer dimensions there are known for generic states and in the extremal qubit case [BRVR18, BLRVR19, LWB08, GKW17], but a genericity statement does not decide a particular state.

Remark B.6 (Two-uniformity does not identify the group). An explicit state shows that the second half of the rigidity phenomenon needs strictly more than the first. Let $q = 2$ and $C = \text{RM}(1, 4) \leq \mathbb{F}_2^{16}$, whose nonzero codeword weights are 8 and 16 and whose dual $\text{RM}(2, 4)$ has minimum distance 4. The equal-phase CSS state of C has label group $C_X \oplus C_Z^\perp$, so by (2.2) a marginal fails to be maximally mixed only when a nonzero word of C or of $C^\perp$ is supported inside it; both distances exceed 2, so the state is 2-uniform and Theorem B.3 makes its symmetry group finite. That group is not Clifford. Writing $T = \text{diag}(1, e^{i\pi/4})$, the product $T^{\otimes 16}$ multiplies $|x\rangle$ by $e^{i\pi \text{wt}(x)/4}$, which is 1 at every codeword weight, so $T^{\otimes 16}$ fixes the state exactly; and $TX(1)T^\dagger = 2^{-1/2}(X(1) + iX(1)Z(1))$ is not a Weyl axis. Two-uniformity buys finiteness and nothing else. Identifying the finite group as local Clifford is what the AME hypothesis and Theorem 1.1 supply, and the next corollary is that identification.

Corollary B.7 (Discrete product-unitary symmetry). Fix a stabilizer AME state $|\psi\rangle$ as in Theorem 1.1, and define the local-factor tuple groups

$$G = \{(U_i) \in U(q)^{2m} : (\bigotimes_i U_i)|\psi\rangle \in \mathbb{C}|\psi\rangle\},$$

$$\tilde{G} = \{((U_i), \pi) \in U(q)^{2m} \rtimes S_{2m} : (\bigotimes_i U_i)\pi|\psi\rangle \in \mathbb{C}|\psi\rangle\},$$

where S_{2m} permutes the local factors. Let $T = (S^1)^{2m}$ be the coordinatewise scalar subgroup and put $\Gamma = G/T$, $\tilde{\Gamma} = \tilde{G}/T$. Then

$$1 \longrightarrow T \longrightarrow G \longrightarrow \Gamma \longrightarrow 1, \quad 1 \longrightarrow T \longrightarrow \tilde{G} \longrightarrow \tilde{\Gamma} \longrightarrow 1$$

are short exact sequences with closed connected kernel and finite discrete quotient; both inclusions of T are continuous. In particular, G and $\tilde{G}$ are finite unions of connected components, each a translate of T . Thus the local-factor automorphism group modulo one-site scalar phases is finite and discrete, and its identity component before quotienting is exactly T . If Π is the subgroup of party permutations realized by $\tilde{G}$, there is a further exact sequence

$$1 \longrightarrow \Gamma \longrightarrow \tilde{\Gamma} \longrightarrow \Pi \longrightarrow 1.$$

This last extension splits exactly when the realized permutations admit a homomorphic choice of projective product-unitary lifts. We use only the finiteness statement here; the extension class is not part of the quantitative argument.

Proof. Every local factor in a tuple from G or $\tilde{G}$ is Clifford. The projective single-qudit Clifford group over a finite field is finite, as is the party-permutation group. The kernel of projectivizing each local factor is exactly the torus of one-site scalar phases. That torus is the identity component, hence is closed; the finite Hausdorff quotient is discrete. Translation identifies every connected component with the scalar torus, so finitely many quotient representatives cover the group. Forgetting the local factors maps $\tilde{\Gamma}$ onto the realized permutation subgroup Π . Its kernel consists exactly of the fixed-party projective automorphisms, which gives the third sequence. A splitting is, by definition, a group-homomorphic right inverse; arbitrary or generator-by-generator lifts do not provide one. $\square$

Discreteness makes the exact symmetries isolated. The same identity (B.1) says how fast the defect grows away from them, with a constant that does not involve the number of parties.

Theorem B.8 (Local stability). Let $|\psi\rangle$ be 2-uniform, let each h_j be traceless Hermitian on $\mathbb{C}^q$, and put

$$U = \bigotimes_{j=1}^n e^{ih_j}, \quad t = \sum_{j=1}^n \|h_j\|_{\text{op}}, \quad D = \left(\sum_{j=1}^n \|h_j\|_F^2 \right)^{1/2}.$$

If $t \leq \frac{1}{2}$ then

$$\frac{D^2}{q} \left(1 - \frac{t}{3} \right) \leq \varepsilon(U)^2 \leq \frac{D^2}{q} \left(1 + \frac{t}{3} \right). \quad (\text{B.3})$$

In particular $D \leq \sqrt{6q/5} \varepsilon(U) < 1.096\sqrt{q} \varepsilon(U)$, and $\varepsilon(U)^2/D^2 \rightarrow q^{-1}$ as $t \rightarrow 0$, uniformly in n .

Proof. Since the sites commute, $U = e^{iM}$ with $M = \sum_j h_j^{(j)}$, and $\|M\|_{\text{op}} \leq t$ by the triangle inequality. Write $\langle \cdot \rangle = \langle \psi | \cdot | \psi \rangle$. Taylor's formula with integral remainder gives $e^{ix} = 1 + ix - x^2/2 + r(x)$ with $|r(x)| \leq |x|^3/6$ for real x . Let $M = \sum_i \lambda_i P_i$ be the spectral decomposition and put $p_i = \langle P_i \rangle \geq 0$, so that $\sum_i p_i = 1$. Then $\langle r(M) \rangle = \sum_i p_i r(\lambda_i)$ and

$$|\langle r(M) \rangle| \leq \frac{1}{6} \sum_i p_i |\lambda_i|^3 \leq \frac{t \langle M^2 \rangle}{6}.$$

By Lemma B.1, $\langle M \rangle = 0$ and $\langle M^2 \rangle = D^2/q$, so $\langle e^{iM} \rangle = 1 - \langle M^2 \rangle/2 + \langle r(M) \rangle$. Also $\langle M^2 \rangle \leq t^2 \leq \frac{1}{4}$, so $1 - \langle M^2 \rangle/2 > 0$ and

$$\left| |\langle e^{iM} \rangle| - \left(1 - \frac{1}{2} \langle M^2 \rangle \right) \right| \leq \frac{t \langle M^2 \rangle}{6},$$

using $|\langle e^{iM} \rangle| \geq \text{Re} \langle e^{iM} \rangle$ for the lower estimate. Since $\varepsilon(U)^2 = 2 - 2|\langle e^{iM} \rangle|$, this is exactly (B.3). At $t \leq \frac{1}{2}$ the left inequality reads $\varepsilon^2 \geq \frac{5}{6}D^2/q$, and the two-sided form gives the limit. $\square$

The constant $\sqrt{6/5} = 1.0954\dots$ is therefore sharp to within about ten percent: the true asymptotic constant is $\sqrt{q}$, and the excess is the cubic Taylor cutoff, not structure.

Remark B.9 (The stability constant is a Fisher metric). Identity (B.1) also explains why the constant does not involve n . For the family $U(s) = \bigotimes_j e^{ish_j} = e^{isM}$, the quantum Fisher information of a pure state is $4 \text{Var}_\psi(M)$ [BC94], so

$$F_Q = \frac{4}{q} \sum_{j=1}^n \|h_j\|_F^2.$$

Splitting the variance of a sum of one-site terms into single-party variances and cross-party covariances is the standard framework for metrologically useful entanglement [HLK⁺12, Tó12]; 2-uniformity is exactly the hypothesis that annihilates every covariance and pins every single-party variance. On traceless local generators — and only there, since (B.1) says nothing about generators that are not sums of one-site terms — the Fisher form is a flat multiple of the Euclidean one. A degenerate direction would be a continuous product symmetry, which Theorem B.3 excludes; an enhanced direction, of the kind a GHZ state supplies, would need the pair correlations that 2-uniformity forbids. The stability constant is the inverse of that flat metric.

The Fisher form is quadratic data, so it governs the constant and says nothing about how large a neighbourhood the quadratic estimate covers. That size is fixed by the higher moments of M , and Theorem B.12 and Proposition B.15 below determine it.

The size of the certified region is controlled by the state's uniformity order, not by the number of parties. Call $|\psi\rangle$ *k-uniform* if every reduced operator on at most k parties is maximally mixed; 2-uniformity is the case $k = 2$, and an $\text{AME}(2m, q)$ state is m -uniform. The hypothesis $t \leq \frac{1}{2}$ in Theorem B.8 pays for a cubic Taylor remainder. Each additional order of uniformity removes one order from that remainder, because the state and the tracial state assign the same value to every power of M below the uniformity order.

Lemma B.10 (Tracial agreement below the uniformity order). Let $|\psi\rangle$ be k -uniform on n qudits. Then $\langle \psi | O | \psi \rangle = q^{-n} \text{Tr } O$ for every operator O supported on at most k parties. In particular, with $M = \sum_j h_j^{(j)}$ as above, $\langle \psi | M^r | \psi \rangle = q^{-n} \text{Tr}(M^r)$ for every $r \leq k$.

Proof. If $O = O_S \otimes I$ with $|S| \leq k$ then $\langle O \rangle = \text{Tr}(\rho_S^\psi O_S) = q^{-|S|} \text{Tr } O_S = q^{-n} \text{Tr } O$. Expanding M^r gives a sum of products of one-site operators touching at most $r \leq k$ sites, so every term is of that shape. $\square$

The tracial state $q^{-n} \text{Tr}$ is the product of the one-site tracial states, and the site summands of M commute, so $q^{-n} \text{Tr } f(M) = \mathbb{E} f(X_1 + \dots + X_n)$ with the X_j independent and X_j uniform on the spectrum of h_j . The tracial main term is therefore a product of one-site characteristic functions, and the following contraction estimate controls each factor.

Lemma B.11 (Per-site contraction). Let h be traceless Hermitian on $\mathbb{C}^q$ with spectral spread $s = \lambda_{\max}(h) - \lambda_{\min}(h)$. Then

$$|q^{-1} \text{Tr } e^{ih}|^2 \leq 1 - c(s) \|h\|_F^2 / q, \quad c(s) = \max \left\{ 1 - \frac{s^2}{12}, \frac{4}{\pi^2} [s \leq \pi] \right\}.$$

Proof. Write $q^{-1} \text{Tr } e^{ih} = \mathbb{E} e^{iX}$ with X uniform on the spectrum. For an independent copy X' , $|\mathbb{E} e^{iX}|^2 = \mathbb{E} \cos(X - X')$ and $|X - X'| \leq s$. From $\cos u \leq 1 - u^2/2 + u^4/24 = 1 - (u^2/2)(1 -$

$u^2/12$) and $u^2 \leq s^2$, $\mathbb{E} \cos(X - X') \leq 1 - (1 - s^2/12) \text{Var } X$; from $\cos u \leq 1 - 2u^2/\pi^2$ for $|u| \leq \pi$, $\mathbb{E} \cos(X - X') \leq 1 - (4/\pi^2) \text{Var } X$. Tracelessness gives $\text{Var } X = \text{Tr}(h^2)/q = \|h\|_F^2/q$. $\square$

Theorem B.12 (Stability at uniformity order k). Let $|\psi\rangle$ be k -uniform with $k \geq 2$, let each h_j be traceless Hermitian with spectral spread at most s , and keep t, D, U as in Theorem B.8. Put

$$c = c(s), \quad \lambda = \frac{2t^{k-1}}{(k+1)!}, \quad \rho = c - \frac{c^2 D^2}{4q} - 2\lambda.$$

Then $\varepsilon(U)^2 \geq \rho D^2/q$; in particular $\rho > 0$ gives $D \leq \sqrt{q/\rho} \varepsilon(U)$.

Proof. Write $e^{ix} = \sum_{r \leq k} (ix)^r/r! + \varrho(x)$ with $|\varrho(x)| \leq |x|^{k+1}/(k+1)!$. By Lemma B.10 the state and the tracial state agree on every M^r with $r \leq k$, so $\langle e^{iM} \rangle - q^{-n} \text{Tr } e^{iM} = \langle \varrho(M) \rangle - q^{-n} \text{Tr } \varrho(M)$. For any spectral weight vector $p_i \geq 0$ with $\sum_i p_i = 1$,

$$\left| \sum_i p_i \varrho(\lambda_i) \right| \leq \frac{\|M\|_{\text{op}}^{k-1}}{(k+1)!} \sum_i p_i \lambda_i^2,$$

and both states give M^2 the value D^2/q , by Lemma B.10 with $r = 2$ on one side and by tracelessness on the other. Since $\|M\|_{\text{op}} \leq t$, each term is at most $t^{k-1} D^2/(q(k+1)!)$, so the two expectations differ by at most $\lambda D^2/q$. The sites commute, so $q^{-n} \text{Tr } e^{iM} = \prod_j q^{-1} \text{Tr } e^{ih_j}$, and Lemma B.11 with $1 - y \leq e^{-y}$ gives

$$|\langle e^{iM} \rangle| \leq \prod_j (1 - c\|h_j\|_F^2/q)^{1/2} + \lambda D^2/q \leq e^{-cD^2/(2q)} + \lambda D^2/q.$$

With $x = cD^2/(2q)$ and $1 - e^{-x} \geq x - x^2/2$, $\varepsilon(U)^2 = 2 - 2|\langle e^{iM} \rangle| \geq 2x - x^2 - 2\lambda D^2/q = \rho D^2/q$. $\square$

Corollary B.13 (The certified radius grows with the uniformity order). Let $|\psi\rangle$ be k -uniform with $k \geq 2$. If every h_j has spectral spread at most $\frac{1}{2}$, if $D \leq \sqrt{q}/2$, and if

$$t \leq R_k := \left(\frac{(k+1)!}{48} \right)^{1/(k-1)},$$

then $D \leq \sqrt{6q/5} \varepsilon(U)$. One has $\log R_k = \log k - 1 + O(\log k/k)$, so $R_k \sim k/e$.

Proof. Spread at most $\frac{1}{2}$ gives $c \geq 1 - \frac{1}{48} = 0.97917$; with $D^2 \leq q/4$, $c^2 D^2/(4q) \leq c^2/16$, so $c - c^2 D^2/(4q) \geq 0.91924$; and $t^{k-1} \leq (k+1)!/48$ gives $2\lambda \leq \frac{1}{12}$. Hence $\rho \geq 0.83591 > \frac{5}{6}$ and $\sqrt{q/\rho} \leq \sqrt{6q/5}$. The asymptotics are Stirling's formula applied to $\log R_k = (\log(k+1)! - \log 48)/(k-1)$. $\square$

At $k = 2$ this is weaker than Theorem B.8, which reaches $t \leq \frac{1}{2}$ while $R_2 = \frac{1}{8}$; the loss is the passage through the tracial main term, which the direct argument avoids. The gain begins at $k = 3$, where $R_3 = 1/\sqrt{2}$, and grows from there: $R_4 = 1.357$, $R_{10} = 4.548$, $R_{30} = 12.93$. For a stabilizer $\text{AME}(2m, q)$ state $k = m$, so the certified radius grows linearly in the party count. It cannot grow faster.

Proposition B.14 (Ceiling on the certified radius). For every $q \geq 2$ and every n there are traceless Hermitian h_j on $\mathbb{C}^q$ with $\sum_j \|h_j\|_{\text{op}} = 2\pi(1 - 1/q)n$ and $D > 0$ for which $\bigotimes_j e^{ih_j}$ is a scalar, hence $\varepsilon = 0$. No estimate $\varepsilon(U)^2 \geq \rho D^2/q$ with $\rho > 0$ can therefore hold on a region of radius exceeding $2\pi(1 - 1/q)n$.

Proof. Take $h_j = (2\pi/q) \text{diag}(q-1, -1, \dots, -1)$ at every site. It is traceless, and both eigenvalues of e^{ih_j} equal $e^{-2\pi i/q}$, so $e^{ih_j} = e^{-2\pi i/q} I$ and the product of n of them is a global phase. Here $\|h_j\|_{\text{op}} = 2\pi(q-1)/q$ and $\|h_j\|_F^2 = (2\pi/q)^2 q(q-1) > 0$. $\square$

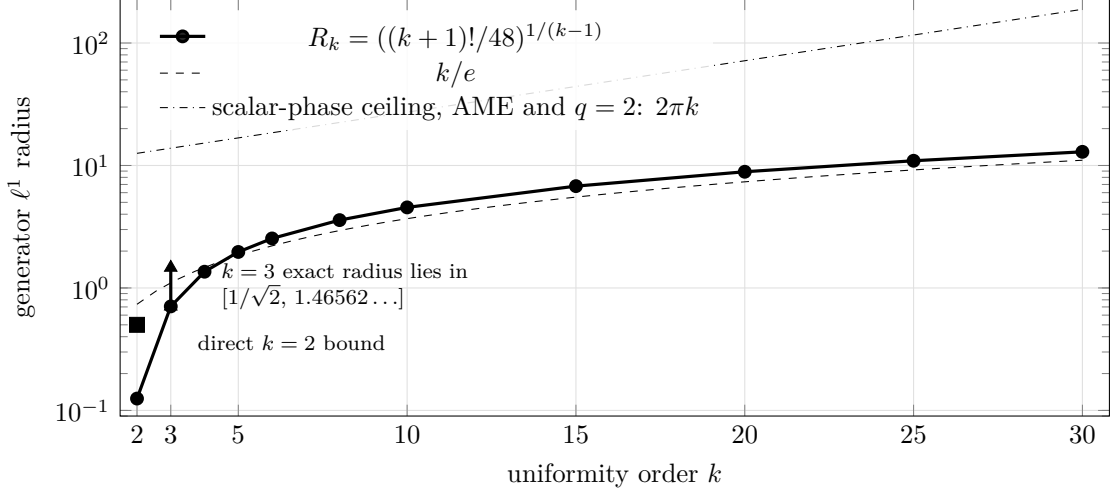


Figure 4: Certified generator ℓ^1 radius of the quadratic stability estimate against uniformity order. The general radius $R_k = ((k+1)!/48)^{1/(k-1)}$ grows as k/e ; scalar-phase configurations bound every such radius by $2\pi(1 - 1/q)n$, so linear growth is the maximal order. For an $\text{AME}(2m, q)$ state, $k = m$. At $k = 3$, the Reed–Muller family pins the exact radius for the constant $\sqrt{6q/5}$ inside the displayed interval.

Linear growth in the party count is therefore the maximal possible order, and Corollary B.13 attains it for AME states, within a factor $4\pi e(1 - 1/q)$ of the ceiling. Bounded uniformity order is a different regime, and the following family measures it exactly.

Proposition B.15 (The certified radius at uniformity order three). Let $\ell \geq 3$, $N = 2^\ell$, and let $|C\rangle$ be the equal-phase CSS state of $C = \text{RM}(1, \ell) \leq \mathbb{F}_2^N$. Its uniformity order is 3, for every ℓ . Take $h_j = cZ(1)$ at every site and put $\theta = Nc \in (0, 2\pi)$. Then

$$\sum_j \|h_j\|_{\text{op}} = \theta, \quad D = \frac{\theta\sqrt{2}}{\sqrt{N}}, \quad \varepsilon = \frac{2|\sin(\theta/2)|}{\sqrt{N}}, \quad \frac{D}{\sqrt{2}\varepsilon} = \frac{\theta}{2\sin(\theta/2)}.$$

The last ratio does not depend on N and increases on $(0, 2\pi)$; it passes $\sqrt{6/5}$ at the unique $\theta^* = 1.46562\dots$. Hence the conclusion of Theorem B.8 fails at $t = \theta^*$, for every length. At uniformity order three the exact radius for the constant $\sqrt{6q/5}$ therefore lies between $R_3 = 1/\sqrt{2}$ and θ^* .

Proof. The nonzero weights of $\text{RM}(1, \ell)$ are $N/2$ and N , and $C^\perp = \text{RM}(\ell-2, \ell)$ has minimum distance 4. A marginal of the equal-phase CSS state fails to be maximally mixed only when a nonzero word of C or of $C^\perp$ is supported inside it, so by (2.2) the uniformity order is $\min(N/2, 4) - 1 = 3$ whenever $N \geq 8$. Since $Z(1)|x\rangle = (-1)^x|x\rangle$ at a single site, $U|x\rangle = e^{ic(N-2\text{wt}(x))}|x\rangle$, and the weight distribution 1, $2N-2$, 1 on weights 0, $N/2$, N gives $\langle U \rangle = 1 - (1 - \cos \theta)/N$. This is nonnegative, so $\varepsilon^2 = 2 - 2\langle U \rangle = 4\sin^2(\theta/2)/N$. Also $\|cZ(1)\|_F^2 = 2c^2$ and $\|cZ(1)\|_{\text{op}} = c$, giving the displayed D and θ . Finally $\theta/(2\sin(\theta/2)) = u/\sin u$ with $u = \theta/2 \in (0, \pi)$, which increases from 1 to ∞ ; at θ^* the ratio exceeds the constant of Theorem B.8 while the defect is $2\sin(\theta^*/2)N^{-1/2}$. $\square$

Two consequences. The smallness hypothesis of Theorem B.8 is load-bearing, and its constant $\frac{1}{2}$ is within a factor of about three of sharp, since the family leaves the hypothesis at $\theta = \frac{1}{2}$ and violates the conclusion at θ^* . And the party count is not what governs the region. The family has N parties and uniformity order three at every length, so what it bounds is the radius at order three; the defect at which it breaks the conclusion is $1.3379\dots N^{-1/2}$

only because a bounded ℓ^1 radius spread evenly over N sites has defect image of order $N^{-1/2}$. When the uniformity order grows with the party count, as it does for an $\text{AME}(2m, q)$ state, the admissible generator radius grows with it; the independent constraint $D \leq \sqrt{q}/2$ then binds before $t \leq R_m \sim m/e$ does. This enlarges a region already described in generator coordinates. It does not imply that every unitary in a constant-radius defect ball has such coordinates. Both statements concern a known state acted on by product unitaries of bounded generator norm; neither certifies an unknown device.

For stabilizer AME states there is a second route to a stability estimate, which buys the same conclusion with a different hypothesis set. It gives up the global ℓ^1 budget on $\sum_j \|h_j\|_{\text{op}}$ entirely, keeps a per-site condition and the global ℓ^2 condition, and pays only in the constant. The mechanism is a bipartition bound in which the Weyl amplitudes of the two halves are compared directly, so no Taylor remainder appears and no budget is needed to control one.

Proposition B.16 (Half-splitting bound). Let $|\psi\rangle$ be a stabilizer $\text{AME}(2m, q)$ state with label group L and character χ , let $U = \bigotimes_j U_j$ be a product unitary, and put $a_j(v) = q^{-1} \text{Tr}(W_v^\dagger U_j)$, so that $p_j = |a_j|^2$ is a probability distribution on $\mathbb{F}_q^2$ for each j . For a set A of m parties define $u, v \in \mathbb{R}^L$ by

$$u(\ell) = \prod_{j \in A} |a_j(\ell_j)|, \quad v(\ell) = \prod_{j \notin A} |a_j(\ell_j)|.$$

Then $\|u\| = \|v\| = 1$ and $|\langle \psi | U | \psi \rangle| \leq \langle u, v \rangle$, so $\varepsilon(U)^2 \geq \|u - v\|^2$.

Proof. The coordinate restriction $\pi_A : L \rightarrow (\mathbb{F}_q^2)^A$ has kernel the labels supported on the complementary m -set, which by (4.1) is trivial; both sides have q^{2m} elements, so π_A is a bijection, and likewise for π_{A^c} . Hence $\|u\|^2 = \sum_{x \in (\mathbb{F}_q^2)^A} \prod_{j \in A} p_j(x_j) = 1$, and the same for v . From (2.2) and $\text{Tr}(W_\ell U) = \prod_j \text{Tr}(W_{\ell_j} U_j)$, taking moduli termwise gives $|\langle \psi | U | \psi \rangle| \leq \sum_{\ell \in L} \prod_j |a_j(\ell_j)| = \langle u, v \rangle$, the relabelling $\ell \mapsto -\ell$ being a bijection of L . For unit vectors $\langle u, v \rangle = 1 - \frac{1}{2} \|u - v\|^2$. $\square$

The AME condition supplies more than the two bijections: it pins down a family of labels of minimum weight that straddles the cut, and those are the labels on which u and v are forced apart.

Lemma B.17 (Cut-transversal labels). With A as above and $j \notin A$, let $L_j = \{\ell \in L : \text{supp } \ell \subseteq A \cup \{j\}\}$. Then $|L_j| = q^2$, every nonzero $\ell \in L_j$ has support exactly $A \cup \{j\}$, and for every $i \in A \cup \{j\}$ the evaluation $\ell \mapsto \ell_i$ is a bijection $L_j \rightarrow \mathbb{F}_q^2$.

Proof. L_j is the kernel of the projection of L to the coordinates $A^c \setminus \{j\}$, which is surjective because it factors the bijection π_{A^c} ; so $|L_j| = q^{2m}/q^{2(m-1)} = q^2$. A nonzero $\ell \in L_j$ has weight at least $m+1$ by (4.1) and support inside a set of that size. For $i \in A \cup \{j\}$ the kernel of $\ell \mapsto \ell_i$ on L_j consists of labels supported on m parties, hence is trivial, and the two sets have the same cardinality. $\square$

Theorem B.18 (Stability without a generator budget). Let $|\psi\rangle$ be a stabilizer $\text{AME}(2m, q)$ state with $m \geq 2$, let $U = \bigotimes_j U_j$ be a product unitary, and put $\eta_j = 1 - |q^{-1} \text{Tr } U_j|^2$ and $H = \sum_j \eta_j$. If $H \leq \frac{1}{4}$ then $\varepsilon(U)^2 \geq H/4$. Consequently, if $U = \bigotimes_j e^{ih_j}$ with each h_j traceless Hermitian of spectral spread at most π and $D \leq \sqrt{q}/2$, then

$$\varepsilon(U)^2 \geq \frac{c D^2}{4q}, \quad \text{that is} \quad D \leq 2\sqrt{q/c} \varepsilon(U),$$

with $c = c(s)$ as in Lemma B.11. No bound on $\sum_j \|h_j\|_{\text{op}}$ is assumed.

Proof. Fix an m -set A and write $\eta = \max_j \eta_j \leq H$. For $j \notin A$ and $w \in \mathbb{F}_q^2$ let $\ell^{j,w}$ be the label of Lemma B.17 with j -coordinate w . Since $\ell^{j,w}$ vanishes off $A \cup \{j\}$,

$$\sum_{w \neq 0} v(\ell^{j,w})^2 = \eta_j \prod_{i \notin A \cup \{j\}} (1 - \eta_i) \geq \eta_j(1 - H).$$

Every A -coordinate of $\ell^{j,w}$ is nonzero, so $p_i(\ell^{j,w}) \leq \eta_i$ for $i \in A$; fixing one $i_0 \in A$ and using injectivity of $w \mapsto \ell_{i_0}^{j,w}$, $\sum_{w \neq 0} u(\ell^{j,w})^2 \leq \prod_{i \in A} \eta_i$. Summing over the m choices of j and using $\prod_{i \in A} \eta_i \leq \eta^{m-1} m^{-1} \sum_{i \in A} \eta_i$ bounds the total by $\eta^{m-1} \sum_{i \in A} \eta_i$. All terms of $\|u - v\|^2$ are nonnegative, and $(x - y)^2 \geq x^2/2 - y^2$, so this family contributes at least $\frac{1}{2}(1 - H) \sum_{j \notin A} \eta_j - \eta^{m-1} \sum_{i \in A} \eta_i$. The mirror family, supported on $A^c \cup \{i\}$ for $i \in A$, is disjoint from it for $m \geq 2$ and contributes the reflected bound. Adding,

$$\varepsilon(U)^2 \geq \|u - v\|^2 \geq \left(\frac{1}{2}(1 - H) - \eta^{m-1}\right)H.$$

For $m \geq 3$, the assumptions give $\eta^{m-1} \leq H^2 \leq \frac{1}{16}$, so the bracket is at least $\frac{5}{16}$. It remains to retain the sharper terms when $m = 2$. Write $A = \{a_1, a_2\}$, $A^c = \{b_1, b_2\}$, and $P = \eta_{a_1}\eta_{a_2} + \eta_{b_1}\eta_{b_2}$. Before the estimates by $1 - H$ and η^{m-1} , the four disjoint minimum-support families above contribute at least

$$\frac{1}{2}(\eta_{a_1}(1 - \eta_{a_2}) + \eta_{a_2}(1 - \eta_{a_1}) + \eta_{b_1}(1 - \eta_{b_2}) + \eta_{b_2}(1 - \eta_{b_1})) - 2P = \frac{H}{2} - 3P.$$

If $s = \eta_{a_1} + \eta_{a_2}$, then

$$P \leq \frac{s^2 + (H - s)^2}{4} \leq \frac{H^2}{4}.$$

Thus at $H \leq \frac{1}{4}$ the last display is at least $(\frac{1}{2} - \frac{3}{16})H = \frac{5}{16}H$. In both cases this is at least $H/4$. For the second statement, Lemma B.11 gives $\eta_j \geq c\|h_j\|_F^2/q$, hence $H \geq cD^2/q$; and $1 - |\mathbb{E} e^{iX}|^2 \leq \frac{1}{2}\mathbb{E}(X - X')^2 = \|h_j\|_F^2/q$ gives $H \leq D^2/q \leq \frac{1}{4}$. $\square$

The two routes are a choice of hypothesis set. Corollary B.13 assumes per-site spread $\frac{1}{2}$, $D \leq \sqrt{q}/2$, and the ℓ^1 budget $t \leq R_k$, and delivers the constant $\sqrt{6q/5}$. Theorem B.18 keeps $D \leq \sqrt{q}/2$, relaxes the spread hypothesis to π , deletes the budget, and pays $2\sqrt{q}/c$, which is $\pi\sqrt{q}$ at spread π and $2.02\sqrt{q}$ at spread $\frac{1}{2}$. Its per-site spread condition is not decorative: the family of Proposition B.14 has e^{ih_j} scalar, so every η_j vanishes and the first bound is empty, and those h_j have spread 2π , exactly where $c(s) \leq 0$ and the second bound is empty too. The spread condition is the residue of the ceiling.

One configuration is worth recording, because it is the natural candidate for a product unitary that Proposition B.16 might fail to see. At $q = 2$ take $U_j = H$ at every site, so that each p_j is uniform on $\{X, Z\}$; then $u(\ell)v(\ell) = 2^{-m}$ when every $\ell_j \in \{X, Z\}$ and vanishes otherwise, so $\langle u, v \rangle = 2^{-m}|L \cap \{X, Z\}^{2m}|$. Writing $\{X, Z\} = X + \{0, Y\}$, the set $L \cap \{X, Z\}^{2m}$ is empty or a coset of $L \cap \{0, Y\}^{2m}$, and the latter is a binary linear code of length $2m$ whose nonzero words have weight at least $m + 1$. Singleton bounds its dimension by m , with equality only for a binary $[2m, m, m + 1]$ MDS code; the binary MDS codes are the trivial ones, which forces $m \leq 1$. Hence $\langle u, v \rangle \leq \frac{1}{2}$ and $\varepsilon(H^{\otimes 2m})^2 \geq 1$ for every stabilizer $\text{AME}(2m, 2)$ state with $m \geq 2$. The bound is not merely nonvacuous there; it is within $\sqrt{2}$ of the largest possible defect. What the half-splitting bound discards is phase information, but the mechanism that detects this configuration is a count of codewords in a structured subset of L , and the MDS constraints force that count down.

Corollary B.19 (Approximate symmetries decompose). Let $|\psi\rangle$ be a stabilizer $\text{AME}(2m, q)$ state with $m \geq 2$. There is an $\varepsilon_0(\psi) > 0$ such that every product unitary U with $\varepsilon(U) < \varepsilon_0$

factors as

$$U = g \cdot \bigotimes_{j=1}^{2m} e^{ih_j}, \quad g \in G(\psi),$$

with the h_j traceless Hermitian, $\sum_j \|h_j\|_{\text{op}} \leq \frac{1}{2}$, and

$$\left(\sum_j \|h_j\|_F^2 \right)^{1/2} \leq \sqrt{6q/5} \varepsilon(U).$$

By Theorem 1.1 the factor g is an exact local-Clifford symmetry.

The size constraint on the h_j is part of the conclusion and cannot be dropped: the generators of Proposition B.14 are traceless with $\bigotimes_j e^{ih_j}$ scalar, so the defect vanishes while the generator norm does not.

Proof. Work on the compact manifold $X = \text{PU}(q)^{2m}$, on which $f(U) = \varepsilon(U)^2$ is well defined because the defect depends on U only through $|\langle \psi | U | \psi \rangle|$.

Quadratic growth. Let $B \subseteq X$ be the image of $\{(h_j) \text{ traceless} : \sum_j \|h_j\|_{\text{op}} \leq \frac{1}{2}\}$ under $(h_j) \mapsto [\bigotimes_j e^{ih_j}]$. That map is a diffeomorphism near the origin, of matching dimension $2m(q^2 - 1)$, so B is a neighbourhood of the identity coset, and Theorem B.8 gives $f \geq \frac{5}{6q} \sum_j \|h_j\|_F^2$ on it. For $g \in G(\psi)$ one has $f(gV) = f(V)$: if $g|\psi\rangle = e^{i\theta}|\psi\rangle$ then $\langle \psi | gV | \psi \rangle = e^{-i\theta} \langle \psi | V | \psi \rangle$ after conjugating the left factor, and f depends only on the modulus. Hence the same estimate holds on the left translate gB , with the generators measured from g , which is the form the factorization $U = g \cdot \bigotimes_j e^{ih_j}$ requires.

Isolation and the threshold. The zero set of f on X is the image Z of $G(\psi)$, finite by Theorem B.3 and Remark B.4. So $\bigcup_{g \in Z} gB$ is a neighbourhood of Z , its complement K is compact, and $f > 0$ on K ; put $\varepsilon_0 = (\min_K f)^{1/2} > 0$. If $\varepsilon(U) < \varepsilon_0$ then $U \notin K$, so $U \in gB$ for some $g \in Z$, and the first step applies at that g . $\square$

The stabilizer hypothesis enters this corollary only in the last sentence of its statement. The factorization itself holds for every 2-uniform state, with g an exact product-unitary symmetry; and finiteness of the zero set is a consequence of (B.1) rather than a separate input, since quadratic growth already isolates the zeros of f on a compact manifold. The threshold this proof produces comes from compactness and is not exhibited; Theorem B.26 below replaces it by a formula, at the cost of the stabilizer hypothesis and of a factor exponential in the party count.

Corollary B.19 measures how far an approximate symmetry of one state is from an exact one. The intertwiner question between two states is different, and its answer is the tool the explicit threshold needs: it converts a small global defect into a nearby Clifford at each party, with a threshold exhibited in closed form.

Lemma B.20 (Quantitative diagonal-tensor axes). Let $r \geq 3$, let $E_1, \dots, E_r$ be N -dimensional complex inner product spaces with orthonormal bases $\{e_{ij}\}$, and put

$$T = a \sum_{j=1}^N \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \quad a > 0, \quad |\lambda_j| = 1.$$

Let T' have the same form in orthonormal bases $\{e'_{ij}\}$ of spaces E'_i , and let $A_i : E_i \rightarrow E'_i$ be unitaries with

$$\|(A_1 \otimes \cdots \otimes A_r)T - T'\| \leq \eta, \quad \delta = \eta/a < 1/\sqrt{2}.$$

Then in each factor there is a unique permutation π_i with

$$\min_{|\alpha|=1} \|A_i e_{ij} - \alpha e'_{i, \pi_i(j)}\| \leq \sqrt{2} \delta \quad (1 \leq j \leq N). \quad (\text{B.4})$$

Proof. Contract both tensors in the first factor against a target frame vector e'_{1k} . The contraction of T' has rank one, and the contraction of $(A_1 \otimes \cdots \otimes A_r)T$ is within η of it. Flattening the latter between its first remaining factor and the last $r - 2$ exhibits its singular values as $a|\langle e'_{1k}, A_1 e_{1j} \rangle|$, $1 \leq j \leq N$. By the Eckart–Young theorem the squared mass outside the largest is at most η^2 , hence at most δ^2 after dividing by a . Unitarity of A_1 and completeness of the source basis make the total squared mass one, so the dominant coefficient has square at least $1 - \delta^2 > \frac{1}{2}$. Two orthogonal rows cannot select the same dominant column, and the overlap matrix is square, so the selections form a permutation π_1 . Each matched column also has total squared mass one and contains that coefficient, so its off-axis mass is at most δ^2 , and $2(1 - \sqrt{1 - \delta^2}) \leq 2\delta^2$ gives (B.4). Repeat in every factor. $\square$

Proposition B.21 (Quantitative intertwiner rigidity). Let $q = p^e$, let $m \geq 2$, and let $|\psi\rangle, |\phi\rangle$ be stabilizer AME($2m, q$) states. Put

$$\tau_p = \min\left\{\frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}}\right\}, \quad \delta = 2q^{(m+1)/2}\varepsilon.$$

Suppose that, after an allowed party permutation,

$$\min_{|\alpha|=1} \|(U_1 \otimes \cdots \otimes U_{2m})|\psi\rangle - \alpha|\phi\rangle\| \leq \varepsilon$$

and that $\delta < \tau_p$. Then for every party i there is a single-qudit Clifford K_i with

$$q^{-1/2}\|U_i - K_i\|_{\text{HS}} \leq \sqrt{2}\delta = 2\sqrt{2}q^{(m+1)/2}\varepsilon. \quad (\text{B.5})$$

For $m = 3$ and equal-phase CSS states of linear $[6, 3, 4]_q$ MDS codes this is the bound $2\sqrt{2}q^2\varepsilon$ under $2q^2\varepsilon < \tau_p$.

Proof. Fix an $(m + 1)$ -party set S . By (2.2) and Proposition 3.4 the supported label group $L_\psi(S)$ has q^2 elements and projects bijectively onto $\mathbb{F}_q^2$ at every party of S , so the nonidentity part of the reduced operator is

$$T_S = q^{-(m+1)} \sum_{v \neq 0} \lambda_v W_{1, L_1 v} \otimes \cdots \otimes W_{m+1, L_{m+1} v}, \quad |\lambda_v| = 1,$$

where v runs over the nonzero labels at one fixed party of S and each L_i is the resulting parametrization of $L_\psi(S)$ at party i . Normalizing every Weyl operator by $q^{-1/2}$ puts T_S in the form of Lemma B.20 with $r = m + 1 \geq 3$, $N = q^2 - 1$, and $a = q^{-(m+1)}q^{(m+1)/2} = q^{-(m+1)/2}$. Two unit vectors at phase-optimized distance ε have density matrices at trace distance at most 2ε ; partial trace contracts the trace norm, and the Hilbert–Schmidt norm is no larger, so every $(m + 1)$ -party marginal satisfies $\eta \leq 2\varepsilon$ and hence $\eta/a \leq 2q^{(m+1)/2}\varepsilon = \delta$. The lemma then matches Weyl axes within $\kappa = \sqrt{2}\delta$.

It remains to turn approximate axes into an exact Clifford, party by party. Extend the matched label permutation by $\pi(0) = 0$; the identity axis is matched exactly. Conjugating a product of two Weyl operators in the two available orders places the axes labelled $\pi(v + w)$ and $\pi(v) + \pi(w)$ within 3κ . Distinct Weyl axes are at phase-minimized distance $\sqrt{2}$, so $\delta < \frac{1}{3}$ forces $\pi(v + w) = \pi(v) + \pi(w)$; the matching was a permutation, so π is $\mathbb{F}_p$ -linear. Conjugation preserves Weyl commutators exactly, replacing the four factors of a commutator by matched axes costs at most 4κ , and distinct p -th root commutator phases are separated by $2\sin(\pi/p)$, so $\delta < \sin(\pi/p)/(2\sqrt{2})$ forces π to preserve the prime-field symplectic form. By (2.0) it is realized by an exact Clifford K_i .

Finally the phases. Put $V = K_i^\dagger U_i$ and expand $V = \sum_x c_x W_x$ with $\sum_x |c_x|^2 = 1$. Axis closeness gives $|q^{-1} \text{Tr}(W_v^\dagger V W_v V^\dagger)| \geq 1 - \kappa^2/2$ for every v , and character orthogonality turns

this into $\sum_x |c_x|^4 \geq (1 - \kappa^2/2)^2$. Hence some $|c_x| \geq 1 - \kappa^2/2$, so V lies within κ of a phase times W_x ; absorbing that displacement into K_i gives (B.5). Every party lies in some $(m+1)$ -set, as in Corollary 3.3. $\square$

The exponent in δ is a feature of the states rather than slack in the estimate. The nonidentity part of an $(m+1)$ -party marginal of an $\text{AME}(2m, q)$ state has Hilbert–Schmidt norm $q^{-(m+1)/2} \sqrt{q^2 - 1}$: the marginal is nearly maximally mixed, and the Weyl signal that exposes the local factors is that small, so a global defect is diluted to that scale before any marginal can read it off.

The closest general framework is the perturbation theory of nearly orthogonally decomposable tensors, which gives dimension-independent bounds without a singular-value gap [AY20]; the argument above is the equal-weight complex specialization needed here, together with the Weyl multiplication, commutator, and implementing-Clifford steps. Its conclusion is distance from the ambient Clifford group and nothing finer. Whenever two states admit an exact intertwiner outside a proposed smaller subgroup, that intertwiner has positive distance from the finite closed subgroup, so taking $\varepsilon = 0$ falsifies every uniform estimate toward it.

Rounding a product unitary to a nearby product Clifford is only half of a decomposition. The other half is deciding whether that Clifford is an exact symmetry, and for stabilizer states it needs no approximation argument at all, because their overlaps take only finitely many values.

Lemma B.22 (Quantized stabilizer overlap). Let $|\psi\rangle, |\phi\rangle$ be stabilizer states on n qudits of local dimension $q = p^e$, with label groups L_ψ, L_ϕ and stabilizer characters χ_ψ, χ_ϕ . Then

$$|\langle\psi|\phi\rangle|^2 = \begin{cases} |L_\psi \cap L_\phi|/q^n, & \text{if } \chi_\psi = \chi_\phi \text{ on } L_\psi \cap L_\phi, \\ 0, & \text{otherwise.} \end{cases}$$

In particular $|\langle\psi|\phi\rangle|^2 \in \{0\} \cup \{p^{-j} : j \geq 0\}$, so $|\langle\psi|\phi\rangle| > p^{-1/2}$ forces $|\phi\rangle \in \mathbb{C}|\psi\rangle$.

Proof. By (2.2), $\rho_\psi = q^{-n} \sum_{\ell \in L_\psi} \chi_\psi(\ell) W_\ell$ and likewise for ϕ . Weyl orthogonality $\text{Tr}(W_\ell^\dagger W_{\ell'}) = q^n \delta_{\ell, \ell'}$ gives

$$|\langle\psi|\phi\rangle|^2 = \text{Tr}(\rho_\psi \rho_\phi) = q^{-n} \sum_{\ell \in L_\psi \cap L_\phi} (\chi_\psi \overline{\chi_\phi})(\ell).$$

The summand is a character of the group $L_\psi \cap L_\phi$, so the sum is $|L_\psi \cap L_\phi|$ when that character is trivial and 0 otherwise. The group has p -power order and $q^n = p^{en}$, which gives the quantization; the value 1 occurs only when $L_\psi = L_\phi$ and the characters agree, that is, when the two states span the same ray. $\square$

The quantization has a consequence about the symmetry group that is stronger, and easier to remember, than any threshold formula. Theorem B.3 makes $G(\psi)$ finite, hence its elements isolated; on the Clifford locus they are isolated by a fixed amount that no parameter of the problem can erode.

Corollary B.23 (Uniform separation of the exact symmetries). Let $|\psi\rangle$ be a stabilizer state on n qudits of local dimension $q = p^e$, and let K be a product Clifford. Then either $K \in G(\psi)$ or

$$\varepsilon(K) \geq (2 - 2p^{-1/2})^{1/2} \geq (2 - \sqrt{2})^{1/2} = 0.7653 \dots$$

Equivalently, if K and K' are product Cliffords in different cosets of $G(\psi)$, then $K^\dagger K'$ has defect at least this much. The bound depends on the characteristic alone: not on the number of parties, not on the state, and not on the local dimension beyond its characteristic.

Proof. $K|\psi\rangle$ is a stabilizer state, so Lemma B.22 applies to the pair $|\psi\rangle, K|\psi\rangle$: either $|\langle\psi|K|\psi\rangle| \leq p^{-1/2}$, whence $\varepsilon(K)^2 = 2 - 2|\langle\psi|K|\psi\rangle| \geq 2 - 2p^{-1/2}$, or the overlap exceeds $p^{-1/2}$ and $K|\psi\rangle \in \mathbb{C}|\psi\rangle$. For two product Cliffords apply this to $K^\dagger K'$, which is again one. Since $p \geq 2$, the bound is least at $p = 2$. $\square$

So on the Clifford locus the defect has no small nonzero values. This is what removes the compactness step from Corollary B.19: a product unitary with small defect is first rounded to a product Clifford by Proposition B.21, and the rounded operator is then an exact symmetry outright, because a near-miss is impossible.

The rounding need not inspect one minimum-support marginal in isolation. All such marginals are projections of orthogonal exact-support sectors of the same global density-matrix difference. Summing their squared errors therefore recovers part of the signal lost by marginalization.

Lemma B.24 (Collective minimum-support energy). Let $\rho = |\psi\rangle\langle\psi|$, $\Delta = U\rho U^\dagger - \rho$, $n = 2m$, and $r = m + 1$. Then

$$\sum_{\substack{A \subseteq [n] \\ |A|=r}} q^r \|\Delta_A\|_{\text{HS}}^2 \leq q^n \|\Delta\|_{\text{HS}}^2 \leq 2q^n \varepsilon(U)^2. \quad (\text{B.7})$$

Writing $\delta_A = q^{r/2} \|\Delta_A\|_{\text{HS}}$ and $N_m = \binom{2m}{m+1}$, there are two r -sets A, B with $A \cup B = [n]$ and

$$\delta_A^2 + \delta_B^2 \leq \frac{4q^n}{N_m} \varepsilon(U)^2. \quad (\text{B.8})$$

Proof. The stabilizer expansion of ρ has no nonidentity component on at most m sites. Local conjugation preserves exact support because it preserves tracelessness. Thus Δ_A , for $|A| = r$, is precisely the partial trace of the exact-support- A component of Δ . If that component is $D_A \otimes I_{A^c}$, then $\Delta_A = q^{n-r} D_A$ and

$$\|D_A \otimes I_{A^c}\|_{\text{HS}}^2 = q^{-(n-r)} \|\Delta_A\|_{\text{HS}}^2.$$

Exact-support subspaces are mutually orthogonal, which gives the first inequality in (B.7). The second follows from $\|\Delta\|_{\text{HS}}^2 = 2(1 - |\langle\psi|U|\psi\rangle|^2) \leq 2\varepsilon(U)^2$.

Join two r -sets when their union is $[n]$. This graph is regular: for fixed A , the complement of a neighbor is any $(m-1)$ -subset of A . Averaging $\delta_A^2 + \delta_B^2$ over its edges gives twice the vertex average. One edge therefore satisfies (B.8). $\square$

The two good sets cover every party, so the quantitative-axis argument can be applied to A and B , using A on their intersection. This gives one product Clifford while retaining collective ℓ^2 control of the local errors. Two-uniformity then converts that control directly to state distance, without a telescoping ℓ^1 estimate.

Theorem B.25 (Aggregate global rounding). Let $|\psi\rangle$ be a stabilizer $\text{AME}(2m, q)$ state with $q = p^e$ and $m \geq 3$. Put

$$N_m = \binom{2m}{m+1}, \quad r = m+1, \quad d_p = (2 - 2p^{-1/2})^{1/2}, \quad (\text{B.9})$$

$$R_{m,q} = \min \left\{ \frac{\tau_p \sqrt{N_m}}{2q^m}, \frac{\sqrt{N_m}}{2q^{m+1/2}}, \frac{\sqrt{N_m}}{2\pi\sqrt{2r}q^m}, \frac{d_p}{1 + \pi\sqrt{2rq^{2m}/N_m}} \right\}.$$

If $\varepsilon(U) < R_{m,q}$, then, up to a global phase,

$$U = g \bigotimes_{i=1}^{2m} e^{ih_i}, \quad g \in G(\psi),$$

where the h_i are traceless Hermitian, have spectral spread at most π , and satisfy

$$\left(\sum_i \|h_i\|_F^2 \right)^{1/2} \leq \pi \sqrt{q} \varepsilon(U). \quad (\text{B.10})$$

Moreover $R_{m,q} = \Theta_q(m^{-3/4}(2/q)^m)$. Thus this route is polynomial in m at $q = 2$, whenever the stated AME state exists, improves the exponential base at $q = 3$, improves the frame-rounding prefactor at $q = 4$, and is superseded by Theorem B.26 for $q > 4$.

Proof. Choose A, B from Lemma B.24. The nonidentity part of each marginal has diagonal Weyl coefficient $q^{-r/2}$, so Lemma B.20 and its additivity, commutator, and phase steps give a Clifford K_i at each party with

$$q^{-1/2} \|U_i - K_i\|_{\text{HS}} \leq \sqrt{2} \delta_i, \quad \sum_i \delta_i^2 \leq r(\delta_A^2 + \delta_B^2) \leq \frac{4rq^n}{N_m} \varepsilon(U)^2, \quad (\text{B.11})$$

where $\delta_i = \delta_A$ on A and $\delta_i = \delta_B$ on $A^c \subset B$. The first clause of (B.9) supplies the axis threshold τ_p .

Choose principal eigenangles of $K_i^\dagger U_i$ after optimizing its scalar phase and subtract their mean, obtaining traceless h_i . Since $|x| \leq (\pi/2)|e^{ix} - 1|$ for $|x| \leq \pi$,

$$D^2 := \sum_i \|h_i\|_F^2 \leq \frac{\pi^2 q}{2} \sum_i \delta_i^2 \leq \frac{2\pi^2 q r q^n}{N_m} \varepsilon(U)^2. \quad (\text{B.12})$$

The second clause of (B.9) puts all eigenangles in an arc of length π ; the third gives $D \leq \sqrt{q}/2$.

Put $K = \bigotimes_i K_i$ and $V = K^\dagger U = \bigotimes_i e^{ih_i}$, up to phase. If $M = \sum_i h_i^{(i)}$, two-uniformity and tracelessness give $\langle M^2 \rangle = D^2/q$. Hence

$$\|(V - I)\psi\|^2 = \langle 2 - 2 \cos M \rangle \leq \langle M^2 \rangle = D^2/q.$$

Together with (B.12), the fourth clause of (B.9) puts the rays of $K\psi$ and ψ less than d_p apart. Corollary B.23 forces $K \in G(\psi)$. Apply Theorem B.18 to the residual V : its constant has $c \geq 4/\pi^2$, giving (B.10).

Finally $N_m \sim 4^m/\sqrt{\pi m}$. The third and fourth clauses of (B.9) have order $m^{-3/4}(2/q)^m$, and the other two are larger by a polynomial factor. Comparison with the $q^{-(m+2)/2}/m$ scale below gives the stated boundary at $q = 4$. $\square$

The theorem separates frame recognition from the choice of residual metric. If the ℓ^1 conclusion of Corollary B.19 is required, the same covering pair and a telescoping estimate give it below

$$\min \left\{ \frac{\tau_p \sqrt{N_m}}{2q^m}, \frac{\sqrt{N_m}}{4\pi \sqrt{2q} \sqrt{(m+1)^2 + (m-1)^2} q^m} \right\}. \quad (\text{B.13})$$

At $q = 2$ this is of order $m^{-5/4}$. The loss relative to (B.9) comes from imposing a global generator ℓ^1 budget after the frames are known; it is not a remaining marginal-dilution loss.

B.1 The single-marginal decomposition threshold

Theorem B.26 (Explicit decomposition threshold). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state with $q = p^e$, $m \geq 2$, $n = 2m$. Put

$$\varepsilon_0 = \min \left\{ \frac{\tau_p}{2q^{(m+1)/2}}, \frac{1}{4\sqrt{2}\pi n q^{(m+2)/2}} \right\}, \quad \tau_p = \min \left\{ \frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}} \right\}.$$

Every product unitary U with $\varepsilon(U) < \varepsilon_0$ satisfies the conclusion of Corollary B.19.

Proof. Write $\varepsilon = \varepsilon(U)$ and $B = 2\sqrt{2}nq^{(m+2)/2}\varepsilon$, so that $\varepsilon < \varepsilon_0$ gives $B < 1/(2\pi)$.

Rounding. Since $2q^{(m+1)/2}\varepsilon < \tau_p$, Proposition B.21 applied with $|\phi\rangle = |\psi\rangle$ and the identity permutation supplies Cliffords K_i with $\|U_i - K_i\|_{\text{op}} \leq \|U_i - K_i\|_{\text{HS}} \leq 2\sqrt{2}q^{(m+2)/2}\varepsilon$. Summing over the n parties gives $\sum_i \|U_i - K_i\|_{\text{op}} \leq B$.

The rounded Clifford is exact. Put $K = \bigotimes_i K_i$, so that $\|U - K\|_{\text{op}} \leq B$ and $|\langle\psi|K|\psi\rangle| \geq 1 - \varepsilon^2/2 - B$. Here $B < 1/(2\pi) < 0.1592$, and $\varepsilon \leq \tau_p/(2q^{(m+1)/2}) \leq 1/(6 \cdot 2^{3/2}) < 0.06$ because $q \geq 2$ and $m \geq 2$, so the right side exceeds $1 - 0.0018 - 0.1592 > 1/\sqrt{2} \geq p^{-1/2}$. By Corollary B.23, $K \in G(\psi)$; set $g = K$.

The residual generators are small. Each factor of $V = g^\dagger U$ satisfies $\min_\varphi \|K_i^\dagger U_i - e^{i\varphi} I\|_{\text{op}} \leq \|U_i - K_i\|_{\text{op}} \leq B < 1$. For a unitary W on $\mathbb{C}^q$ with $\sigma = \min_\varphi \|W - e^{i\varphi} I\|_{\text{op}} \leq 1$, the eigenvalues of $e^{-i\varphi} W$ at the optimal phase are $e^{i\mu_a}$ with $2|\sin(\mu_a/2)| \leq \sigma$ and $\mu_a \in (-\pi, \pi]$, so $|\mu_a| \leq 2\arcsin(\sigma/2) \leq \pi\sigma/2$; subtracting the mean gives $W = e^{i\varphi'} e^{ih}$ with h traceless Hermitian and $\|h\|_{\text{op}} \leq 2\max_a |\mu_a| \leq \pi\sigma$. Applying this at each site, $\sum_i \|h_i\|_{\text{op}} \leq \pi B < \frac{1}{2}$, and the collected phases are a global phase absorbed into g . Finally $\varepsilon(V) = \varepsilon(U)$, because g fixes the ray of $|\psi\rangle$, so Theorem B.8 applies to V on its hypothesis and gives $D \leq \sqrt{6q/5}\varepsilon(U)$. $\square$

Nothing in this proof is a compactness argument: the finite zero set is never enumerated and no minimum over a compact remainder is taken. What compactness supplied in Corollary B.19 is supplied here by Corollary B.23, in closed form. The factorization is also constructive in the sense that matters: g is produced from U by rounding each local factor to a Clifford, and an $(m+1)$ -party marginal certifies the rounding.

For this single-marginal route the price is the size of ε_0 , which is $q^{-(n+4)/4}/n$ up to constants and therefore exponentially small in the number of parties. Two steps pay it. Rounding costs $q^{(m+1)/2}$, and by the marginal-dilution remark after Proposition B.21 no detector that reads a single small marginal can avoid that factor. Summing n per-site operator-norm errors and taking n logarithms costs a further $n\sqrt{q}$, which is the price of a per-site conclusion.

The exponential gap left by this theorem and Theorem B.25 is closed by Theorem 5.3. The remaining comparison is with Proposition B.15. That proposition bounds any threshold valid across states of uniformity order three; an $\text{AME}(2m, q)$ state has uniformity order m , and for such states Corollary B.13 certifies the quadratic estimate on a generator-coordinate region whose ℓ^1 radius grows with m , while Theorem B.18 removes that budget for every $m \geq 2$. Those estimates control defect from specified generators; they do not put a defect ball inside that coordinate region. The cleaning theorem supplies the inverse statement at radius of order $n^{-1/2}$, with no claim that this party-count dependence is optimal.

B.2 Local gauge groups of two-unitary gates

Corollary B.27 (Local gauge groups of 2-unitary gates). Let W be a two-qudit gate whose associated four-party state $\psi_{ijkl} = q^{-1}W_{kl,ij}$ is $\text{AME}(4, q)$ — equivalently, W is 2-unitary, remaining unitary after realignment and after partial transposition — and let $G(\psi)$ be its product-unitary symmetry group. Then $(u_1, u_2, u_3, u_4) \mapsto (\bar{u}_1, \bar{u}_2, u_3, u_4)$ is an isometric group isomorphism carrying $G(\psi)$ onto the local gauge group

$$\{(v_1 \otimes v_2, u_3 \otimes u_4) : (u_3 \otimes u_4)W(v_1 \otimes v_2)^\dagger = e^{i\varphi}W\},$$

and it preserves $(\sum_j \|h_j\|_F^2)^{1/2}$. Consequently the local gauge group of any 2-unitary gate is finite modulo the local phases, Theorem B.8 transports with its hypothesis $\sum_j \|h_j\|_{\text{op}} \leq \frac{1}{2}$, and Corollary B.19 transports with its threshold; for a gate arising from a stabilizer $\text{AME}(4, q)$ state Theorem B.26 exhibits that threshold at $m = 2$, namely $\varepsilon_0 = \min\{\tau_p/(2q^{3/2}), 1/(16\sqrt{2}\pi q^2)\}$. Gate defects are measured in the metric $q^{-1}\|\cdot\|_F$. For a gate arising from a stabilizer $\text{AME}(4, q)$ state the exact gauge group is local Clifford.

Proof. Normalization is consistent: $\|W\|_F^2 = q^2$ for a unitary on $\mathbb{C}^q \otimes \mathbb{C}^q$, so $\langle \psi | \psi \rangle = \|W\|_F^2 / q^2 = 1$. A product unitary acts on the reshaped gate by $W \mapsto (u_3 \otimes u_4)W(u_1 \otimes u_2)^T$, so $\|U|\psi\rangle - e^{i\theta}|\psi\rangle\| = q^{-1}\|(u_3 \otimes u_4)W(u_1 \otimes u_2)^T - e^{i\theta}W\|_F$; this is the stated metric. Transposition preserves the Frobenius norm and $u \mapsto \bar{u}$ is an automorphism of $U(q)$ fixing $\|h\|_F$ for the corresponding generators, and $(u_1 \otimes u_2)^T = (v_1 \otimes v_2)^\dagger$ with $v_a = \bar{u}_a$, which gives the displayed isomorphism. An $\text{AME}(4, q)$ state is 2-uniform, so Theorem B.3 applies, with Remark B.4 identifying the quotient as the one by the local phases; the stabilizer case is Theorem 1.1. $\square$

The statement is vacuous at $q = 2$, where no $\text{AME}(4, 2)$ state exists. In the dual-unitary circuit literature the local gauge freedom is the standing nuisance parameter of the classification programme [RAL22], and the finite-field Clifford subfamily has been classified, with perfect-tensor cores appearing among its equivalence classes [Pah26]; the corollary says that for 2-unitary gates the gauge group itself is finite and, in the stabilizer case, Clifford.

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